NESTED-SPACE COSMOLOGY

The Geometry of a Universe

Regions, boundaries and what an interior observer can learn

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Abstract

Nested-Space Cosmology asks how a region's geometry, fields and surrounding structure determine the world available to an observer inside it. We develop this question as a finite atlas: a parent with several black-hole boundaries, possible unequal children and further generations. Local spherical geometry provides tractable boundaries and observer charts; it does not make a multi-hole parent globally spherical or establish the topology of our universe. Published spacetime solutions, junction laws, branching mathematics and regional reduction supply the foundations. The work then concentrates on missing connections between them.

One connection is explicit. A fixed de Sitter/Schwarzschild wall joins a regular expanding cap to a vacuum bridge and an asymptotic end on a checked spacelike slice. A vacuum-end gluing theorem supplies a conditional multi-body initial-data family preserving that body. In the isolated spherical reference, the center, moving wall and right exterior have different causal access: selected early signals connect them, while new emissions after the chosen initial slice do not. A prescribed early conformal scalar pulse crosses the wall and produces a calculable transient at the center. Its waveform alone does not establish a unique exterior: the incident state is an independent input. We now identify a finite characteristic preparation in the retained white-hole sector and a bounded exterior-to-wall transfer operator. Its existence is certified analytically. Evaluating it predicts a central pulse followed by a small positive scattering tail within the declared finite window. An independent positive-kernel argument fixes the tail's sign; spatial and clock refinements test its numerical magnitude. For this fixed geometry, a complete compatible signed central history also determines a unique preparation. Positivity and a prescribed source cutoff exclude some histories; geometry comparison requires those input restrictions to be justified independently. At quadratic order the canonically normalized probe adds no singular surface stress, tangential flux jump or normal-stress jump at the fixed wall. With fixed corner geometry and null normalization, the same preparation requires a calculable correction to the initial areal radius and its affine slope. Transverse constraint transport then determines a geometric mass response on that initial null surface, including a negative quadratic coefficient after the compact input. The existing wall field traces also close an analytic exterior-side mass functional through both null fluxes and the full conformal transformation. Retaining the cap's radius-dependent vacuum background at the actual radius now supplies a compatible cap mass functional and a regular, quiet-anchored wall displacement equation. Two-sided mass transport also preserves the temporal and normal junction conditions. The same prepared field now drives a computed quadratic exterior radius and lapse response on a finite characteristic rectangle, checked against independent mass transport and null constraints. Extending the same field evaluator to the retained wedge now supplies a second mass path, the computed wall displacement and its exterior null-clock correction, with independent numerical refinements. Radial reconstruction now also supplies the smooth cap metric with a regular centre and fixed central proper clock. Its mass agrees with the independent wall transport, and inward null integration gives a central arrival correction at fixed exterior input label. The next-order scalar equation now supplies a computed exterior correction with separately measured forcing-cache, spatial-grid and ODE sensitivities. Reflected cap transport and the observer-clock pullback now reconstruct the cubic central scalar coefficient at five interior arrival times, with separate radial and incoming-ray comparisons. A six-case nonlinear calculation now completes the original pre-arrival window with a common cap, wall and exterior. Carrying the incoming cap pulse analytically substantially reduces its transport error. Geometry and clock responses change little under the tested refinements, but scalar time sensitivity and pointwise constraint defects remain unresolved. A fixed-input SSPRK3 comparison reduces the scalar's 256/512-step change to 1.68% of the fine paired peak, while exposing the role of computational profile-knot phase in temporal quadrature. A subsequent comparison at fixed resolution changes the exterior departure derivative. It reduces the root-mean-square incoming constraint and the final wall kinematic mismatch, while slightly increasing the maximum incoming constraint and worsening independent native wall-curvature estimates. The paired wall scalar changes by 0.524% of its new peak. This is a finite numerical comparison, not a converged spacetime or a finite-amplitude remainder bound. Separately, a local characteristic construction and continuation argument now establish a compatible finite-amplitude cap, wall and exterior on a short original source cell, uniformly over a nonzero coupling interval. Certified departure bounds apply to that solution. The regular core remains quiet there. A

common-domain parameter argument gives a quadratic amplitude remainder. Explicit uniform geometric bounds now prove negative sourced corrections to both the material wall radius and the centre-anchored clock at the original onset endpoint, throughout the admitted positive-amplitude range. The exact frame conversion and source-label event shift are retained. Later central arrival and numerical approximation error remain separate. A separate finite spherical field model demonstrates regional transfer, geometric feedback and reduced memory, without identifying its mode-space complement with a parent universe.

These results connect known mathematics in stated domains. They establish neither regular black-hole formation of children nor a complete cosmology. Physical inheritance of constants, quantum-state selection, persistent gravitational effects, antimatter asymmetry and an arrow of time remain open. The atlas makes those gaps explicit and identifies calculations that can resolve them without requiring every microscopic question to be solved first.

I. The question and the spherical idea

1. A universe viewed from within

An observer measures clocks, fields, curvature and correlations within a limited domain. Such measurements need not uniquely identify the larger structure containing that domain. Our question is whether this ambiguity can become a constructive research programme: assemble possible surroundings, determine which attachments satisfy their equations, and ask what each permits an interior observer to experience.

The organizing sequence is **borrow established mathematics, connect compatible regions, investigate branching and inheritance, calculate local consequences, then compare with observations**. A universal action tying together all fields and regions remains an ambitious candidate. It is not a prerequisite for learning whether a particular boundary can be joined or whether a signal can reach a particular observer.

A mature atlas would let a reader follow a possible universe from its surroundings, through its boundaries, into an observer's world and onward through compatible transitions. It would compare families of histories with explicit geometry, dynamics and observables. Completion means dependable coverage of declared model families, including exclusions and observational degeneracies. Establishing that a family describes our universe is a further empirical achievement. The present finite connections are steps toward that destination, not a claim to have reached it.

The first motif contains two spatially separated black holes in a common parent, possible unequal children, and one further generation. A separate motif lets one hole have multiple offspring. These pictures suggest questions; their arrows do not supply an equation of motion, a formation mechanism or a communication channel. Figure 1 distinguishes the broad motif from the specific attachment whose mathematics has been checked.

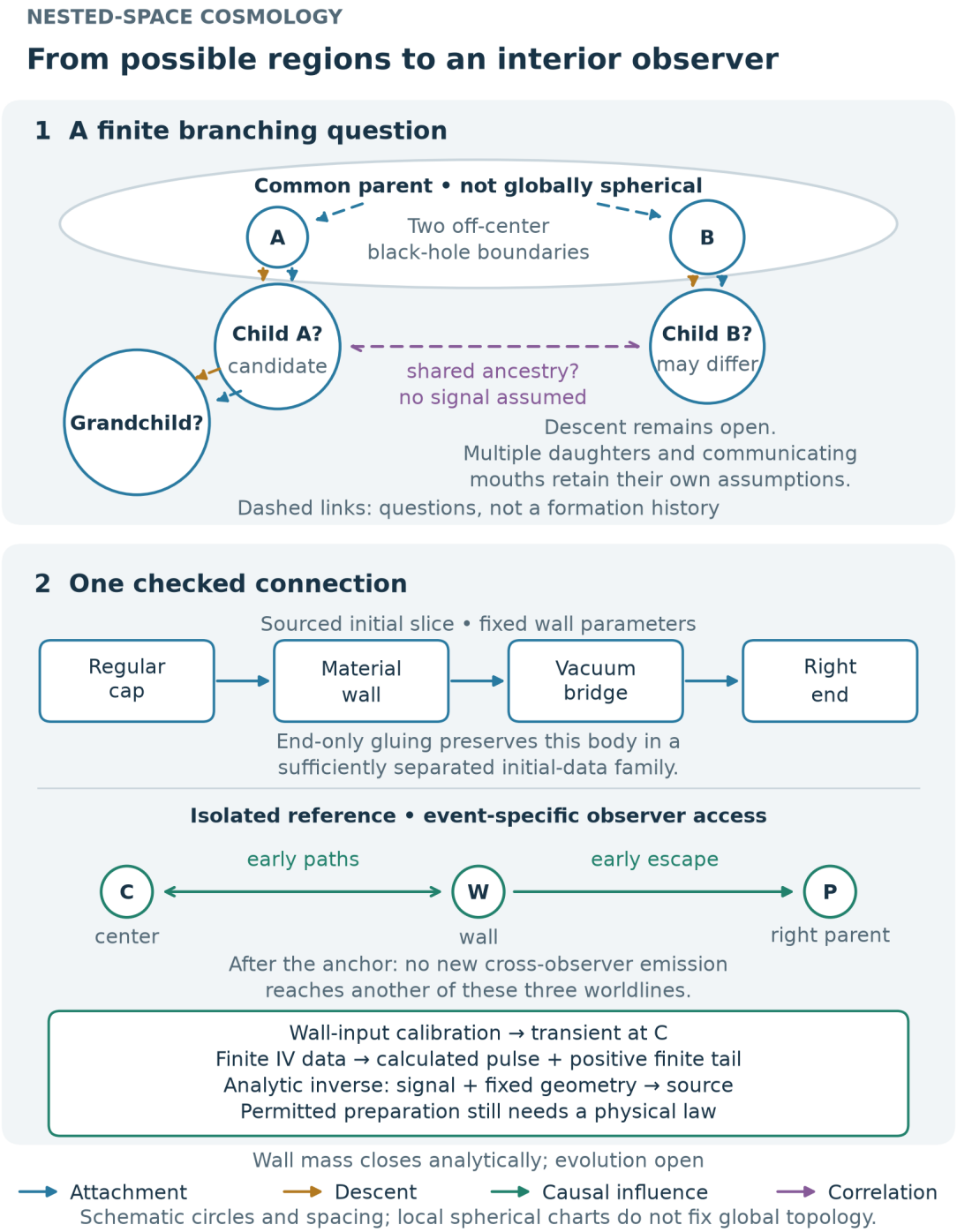


Figure 1. Possible descent (upper panel) remains hypothetical; the lower panel shows the checked cap–wall–bridge body and observer/field reference. Colors distinguish attachment, descent, influence and correlation. Circles and spacing do not establish an embedding, relative scales or global spherical topology. The figure record identifies each connection's evidence and status.

2. What spherical geometry means here

There is a useful borrowed energy argument with a precise domain: in Euclidean space, a ball minimizes boundary area at fixed enclosed volume. A positive constant isotropic surface tension therefore favors a spherical boundary when surface energy is the quantity being minimized. Direction-dependent tension instead selects a Wulff shape; see Neumayer, 1503.06705v2, Section 1.1. This motivates testing rotational symmetry. It does not prove a cosmological equilibrium, dynamical relaxation or a probability ranking over all geometries. Gravity, rotation, matter stresses and the actual boundary law must be specified before importing that variational conclusion into a spacetime model.

A spherical boundary is a two-dimensional surface with area $4\pi R^2$. A spatial interior may be a three-ball; a closed spatial universe may instead have three-sphere topology. These are different assertions. Drawing one circle inside another establishes none of them.

In a locally spherically symmetric spacetime, a useful form is

$$ds^2 = g_{ab}(x)\,dx^a dx^b + R(x)^2 d\Omega^2.$$

Here g_{ab} is a Lorentzian metric on the two-dimensional orbit space and R is areal radius. We use signature *(-+++)* for the atlas. Radial proper distance, areal radius and a diagram's vertical coordinate need not coincide. In particular, a radial profile is not automatically a surface that can be embedded in Euclidean three-space.

The physical spherical hypothesis is that useful cosmological regions or interfaces may admit this structure, exactly or as a controlled approximation. It is not a theorem inferred from statistical isotropy of observations. Several off-center holes generally break global spherical symmetry. We therefore use local spherical charts where justified and an explicitly nonspherical parent where required. Deviations from symmetry are a question for each model, rather than a reason to discard the organizing idea.

3. One thread through three bodies of work

The continuing question is: **how do a region's fields, boundary and surroundings shape its state, geometry and measurements?**

Body of work	What it contributes	Connection still missing
Preserved foundation	Spectral inheritance proposal, boundary operators, source and response constructions	A physical inheritance law and common cosmological source completion
Finite spherical companion	A finite realization of regional transfer, geometric feedback and memory with declared action and state	Identification with an actual assembled parent–child spacetime
Atlas evidence and this living manuscript	Geometric compatibility, a sourced slice, observer access, a calibration response and an evaluated finite exterior preparation	Remote preparation, later evolution, other attachments and observational discrimination

Their mathematics is complementary. A finite operator's eliminated modes are not automatically the exterior spacetime beyond a black hole. The atlas must first identify a field equation, state, clock and matching law before an operator reduction can acquire that geometric meaning.

The foundation's useful historical calculations remain in their source-bound edition and owning notes. This manuscript changes the organizing story: geometry and observer access lead; stronger unifying claims follow only when their missing connections are supplied. No priority claim is made for junction conditions, Schur complements, branching processes or the published wall solutions used below.

II. Regions, inheritance and the meaning of a connection

4. Four arrows that must not be confused

Each region in the atlas carries a geometry, source or state, clock convention, dimensionless parameters and validity domain. Each connection states the borrowed mathematics and the additional condition still needed.

Connection	Mathematical question	What it does not establish by itself
Descent	Where do a child's initial data originate?	A traversable path
Geometric attachment	Which boundaries share admissible metric and source data?	Formation or stability
Causal influence	Can an intervention reach an observer within the stated domain?	A nonzero field response without state and coupling
Correlation	Which information is shared through state or ancestry?	Communication between siblings

Two regions may inherit correlated data without ever exchanging a later signal. An attached interior can lie beyond the causal reach of a specified parent observer. Conversely, communicating mouths require their own causal and material assumptions; they cannot be inserted into a disconnected offspring model as an unlabeled extra arrow.

The multiple-daughter geometry of Barrabès and Frolov (hep-th/9511136v1), the charged semiclassical multi-mouth route of Emparan et al. (2012.07821v2), and quantum-state coupling distributions discussed by Giddings and Turiaci (2004.02900v2) remain separate library entries. A state's coupling distribution is not itself a signal exchanged between its branches. Their source cards and unresolved connections are recorded in the mathematical library.

5. A precise finite inheritance construction

Evidence: exact finite record, frozen snapshot, and proof and domain.

The foundation proposes an inherited normalized law, $T_\Theta^* D_\Theta = D_\Theta$. One controlled meaning is available without postulating an infinite cosmos. Fix a Hermitian d-dimensional matrix H, a nonzero link B, a positive scale ratio Ω, and a finite depth N. For regional labels $j=0,\ldots,N-1$, define a block Hermitian matrix by the following entries where their indices exist:

$$(J_{m,N})_{jj} = \Omega^{m+j} H$$
$$(J_{m,N})_{j,j+1} = \Omega^{m+j} B.$$

The lower off-diagonal blocks are the adjoints. Let U_m consistently relabel the equal-sized regional frames. Then

$$J_{m+1,N} = \Omega U_m J_{m,N} U_m^\dagger$$

$$\Omega^{-1} U_m^\dagger J_{m+1,N} U_m = J_{m,N}.$$

Each block acquires the same extra factor, so the identity follows directly. For fixed depth,

$$\operatorname{spec}(\Omega^{-m} J_{m,N}) = \operatorname{spec}(J_{0,N}).$$

This is a finite realization of normalized inheritance under chosen scaling data. It does not derive Ω, establish a physical transition through a horizon, or show that the universe selects this law. Changing N changes the boundary problem and can change the dressed spectrum. The finite identity is not a proof of cosmic natural selection.

The shared law also does not fix identical states. A fermionic covariance $0 \le C \le I$ evolves as

$$C(t) = e^{-ijt}C(0)e^{ijt}.$$

Regional occupations and cross correlations can differ under the same J. Thus “same laws,” “same state,” and “same observed constants” require separate arguments. A physical transition would have to determine the relevant state and dimensionless data, rather than obtaining them from the normalization identity alone.

6. What an unresolved surrounding region can leave locally

Split a specified operator into retained and complementary blocks A,D with link B. For an admissible complex spectral parameter z, elimination gives the usual self-energy

$$\Sigma(z) = B(zI - D)^{-1}B^{\dagger}.$$

The retained resolvent is $[zI-A-\Sigma(z)]^{-1}$. Nested elimination is associative when the required inverses exist. For the finite inherited chain its normalized recurrence is

$$\Gamma_j(x) = xI - H - \Omega^{-1}B\Gamma_{j+1}(x/\Omega)^{-1}B^{\dagger}.$$

The factor $1/\Omega$ is essential. This is borrowed finite linear algebra applied to the declared family, not an additional force law. To interpret Σ as a cosmological exterior response requires a physical operator, field state and boundary map. Sections 14–15 give a separate finite Hilbert-space realization of this reduction, without identifying its complementary modes with a cosmological exterior.

6.1. The local effective Lambda target

The intended cosmological target is a local effective Λ arising from inherited Schur memory: surrounding structure that a region does not resolve contributes through $\Sigma(z)=B(zI-D)^{-1}B^{\dagger}$ to its local response. A vacuum-form local gravitational part would be written $T^{eff}{}_{\mu\nu}=-(\Lambda_{eff}/\kappa_4)g_{\mu\nu}$, with $\kappa_4=8\pi G$. Identifying the physical exterior operator, its state and boundary map, and obtaining that local part from the causal response remain the missing connections. The finite Schur identity fixes the reduction; it does not yet provide a measured cosmological value. The Schur-memory owner keeps this target and its retained spectral and geometric dictionary.

6.1.1. The current scalar's classical vacuum projection

The current cap-wall candidate allows a direct first calculation of this target. Its action fixes a real massless conformal scalar and independent regional vacuum terms. Contracting the complete improved scalar stress gives

$$T^{\phi\mu}{}_{\mu} = \phi(\Box - \mathcal{R}_J/6)\phi = 0$$
$$\mathcal{R}_J = 4\Lambda_0.$$

Consequently its additional smooth classical trace projection is $\Delta\Lambda_{\rm classical}L_{\rm ref}^{2}=0$, including nonzero on-shell pulses. The scalar still carries stress, mass and flux. The cap baseline $\Lambda_{0,-}\ell^2=3(H\ell)^2=3$ is a supplied input, while $\Lambda_{0,+}=0$ outside. The scalar calculation does not derive that baseline. Distributional wall stress and geometric averaging are separate operations.

The exact calculation record binds the symbolic expressions and their producer. The calculation and frame check also recover the same Jordan curvature from the Einstein-frame potential. Independent metric-jet checks evaluate the implemented nonlinear equations at zero and nonzero amplitude. The scientific owner gives the complete derivation.

6.1.2. A conditional quantum contribution and its physical scale

As an explicit extension, import one quantized real four-dimensional massless conformal scalar with a Hadamard bulk state and locally covariant renormalization. This uses quantum field theory; it does not derive quantum statistics from the atlas. The local trace coefficient follows from the scalar heat-kernel formula, equation (4.28), and conformal anomaly, equations (15)-(16). Substituting $E=-R/6$ cancels its R^2 term. On either smooth quiet region,

$$\langle T^{\mu}{}_{\mu}\rangle = \frac{\hbar}{2880\pi^2}(\mathcal{R}^2_{\mu\nu\rho\sigma} - \mathcal{R}^2_{\mu\nu}).$$

The scheme-dependent $\Box R$ term vanishes because the quiet scalar curvatures are constant. Finite volume and Newton counterterms remain supplied renormalized coefficients. Direct curvature contractions give $(Riemann^2, Ricci^2)=(24H^4, 36H^4)$ in the cap and $(48M^2/R^6, 0)$ outside.

Set $h=H\ell$, $m=M/\ell$, $r=R/\ell$ and $\zeta=\kappa_4\hbar/\ell^2$. With $\Delta\Lambda_{\mathrm{tr}}=-\kappa_4\langle T^{\mu}{}_{\mu}\rangle/4$,

$$\Delta\Lambda_{\mathrm{tr},-}\ell^2 = \frac{\zeta h^4}{960\pi^2}$$
$$\Delta\Lambda_{\mathrm{tr},+}\ell^2 = -\frac{\zeta m^2}{240\pi^2 r^6}.$$

At the supplied $h=1$, $m=2/5$, $r=11/20$, the coefficients of ζ are 0.0001055428996274352 and $-0.0024402417804070 79$. For the independently declared P001 fiducial $\ell=10000\ m$, restoring SI units gives $\zeta=8\pi G\hbar/(c^3\ell^2)$ and

Quiet trace projection	$\Delta\Lambda_{\mathrm{tr}}\ell^2$	Standard uncertainty from G alone
Cap	6.92929×10^{-81}	1.5573×10^{-85}
Exterior at the reference wall radius	-1.60211×10^{-79}	3.6006×10^{-84}

These use CODATA 2022 constants. The length is a fiducial input; the quoted uncertainty is not a total theoretical or observational error band. The exterior term varies with radius. A constant cap trace alone does not establish a stress proportional to its metric, since the material wall need not preserve state symmetry.

There is a necessary test of whether this term can generate the cap scale without supplying a volume term. Put $X=(H\ell)^2$ and $\lambda_0=\Lambda_0\ell^2$. The semiclassical constant-curvature trace equation requires

$$3X = \lambda_0 + \frac{\zeta X^2}{960\pi^2}.$$

At $\lambda_0=0$ its roots are $X=0$ and $X=2880\pi^2/\zeta$. The nonzero root has $H^2\ell_P{}^2=360\pi$, outside the controlled small-curvature semiclassical regime. Thus this one-scalar contribution does not generate a nonzero small-curvature de Sitter cap by itself. The same exact record retains the coefficients, SI inputs and necessary trace condition. No cosmological observation selected these values.

6.1.3. The quantum interface and the remaining exterior reduction

The wall contributes its own local operator data. The existing scalar matching $[\varphi]=0$, $[n\varphi]=k\varphi$ identifies the transmittal coefficient $U_\Sigma=k$. In the inward-normal convention of the transmittal heat-kernel theorem, $L^+=-K_+$ and $L^-=K_-$. Their sum is kI, so the second surface coefficient is proportional to $tr(kI)/3-k=0$.

Keeping every curvature term in the fourth coefficient gives the local surface polynomial

$$a_4^{\Sigma}(1) = \int_{\Sigma} \sqrt{|\gamma|}\, q_{\Sigma}\, d^3y$$
$$q_{\Sigma}\ell^3 = -\frac{b(b^2+3h^2)}{1440\pi^2} = -\frac{7}{720\pi^2}, \quad b = k\ell = 2$$

at $h=1$. This evaluates the imported Riemannian polynomial on the quiet wall's Lorentzian contractions. A joint Lorentzian quantum state and its interface stress have not been constructed. The Gauss and Euler-boundary checks agree with the recorded coefficient. It is a surface invariant; obtaining its metric variation requires the full off-shell operators and their renormalized coefficients.

The physical Schur connection must use the exterior conformal wave operator with this transmission law. For boundary field $\varphi=\varphi|_{\Sigma}$, let $N_{\pm}$ map it to the corresponding outward normal derivative. The homogeneous matching equation is $(N_-+N_++U_{\Sigma})\varphi=0$; incoming initial data add a drive. In real time these are causal history operators. Their retarded response and the joint state's symmetric two-point function are needed for the effective stress. A coefficient multiplying a boundary field quadratic form alone does not supply a metric volume term.

6.2. The retained local incoming gate

Evidence: majorant v7 record and gate budget v5 record.

The local incoming source-insertion majorant v7 is a separate controlled application. For the fixed switched state law $C_{\Sigma}[g]=U_g C_{up} U_g^{\dagger}$ on $I=S(1)+[.12,.18]$, family $(14,1)$ covers **1168/1168 signed rows**, with original weights counted once and no remaining row keys. Its continuous insertion bounds are

$$\varepsilon_N \le 3646674054457366029024452496769420151764188105426118876609\, 2^{-229}.$$

$$\varepsilon_{\beta} \le 5793343250846686183515013928067141454707331929528967339969\, 2^{-230}.$$

These are approximately $4.226958072607968\times10^{-12}$ and $3.357610065481873\times10^{-12}$ for this family's source-error insertion. The physical local gate remains **OPEN**: `physical_upstream_budget_component` is null. Source quadrature, numerical field error, value-assembly arithmetic and every other source family are excluded from these bounds. Complete row coverage in this family does not supply those missing errors.

The v5 gate budget names nine distinct components, with total error allocation 2×10^{-11} and residual reserve 10^{-11} against the physical tolerance 3×10^{-11}. Its applicable component bounds remain null; 60 positive and 120 signed families are declared requirements rather than authenticated coverage. The gate is neither an existence nor a non-existence certificate and does not determine Λ_{eff}. Its paused campaign does not invalidate the finite inheritance theorem or the scalar-chain results. Retired supporting records are cited at Zenodo.

III. A finite map of possible geometries

7. Start with two holes, without manufacturing two children

The explicit parent benchmark uses time-symmetric Brill–Lindquist vacuum initial data,

$$h_{ij} = \psi^4 \delta_{ij}$$
$$K_{ij} = 0$$
$$\psi = 1 + \frac{m_A}{2r_A} + \frac{m_B}{2r_B}.$$

The recorded choice is m_A=1, m_B=1.5, separation d=12. It has outer ADM mass 2.5 and two inner-end masses 1.0625 and 1.5625. These are three asymptotically Euclidean ends of an initial slice. Extra ends do not by themselves become cosmological children. The companion hole supplies nonspherical structure, so an exact isolated Schwarzschild collar cannot simply be assumed at either puncture. The parent record and owning analysis retain the construction and checks.

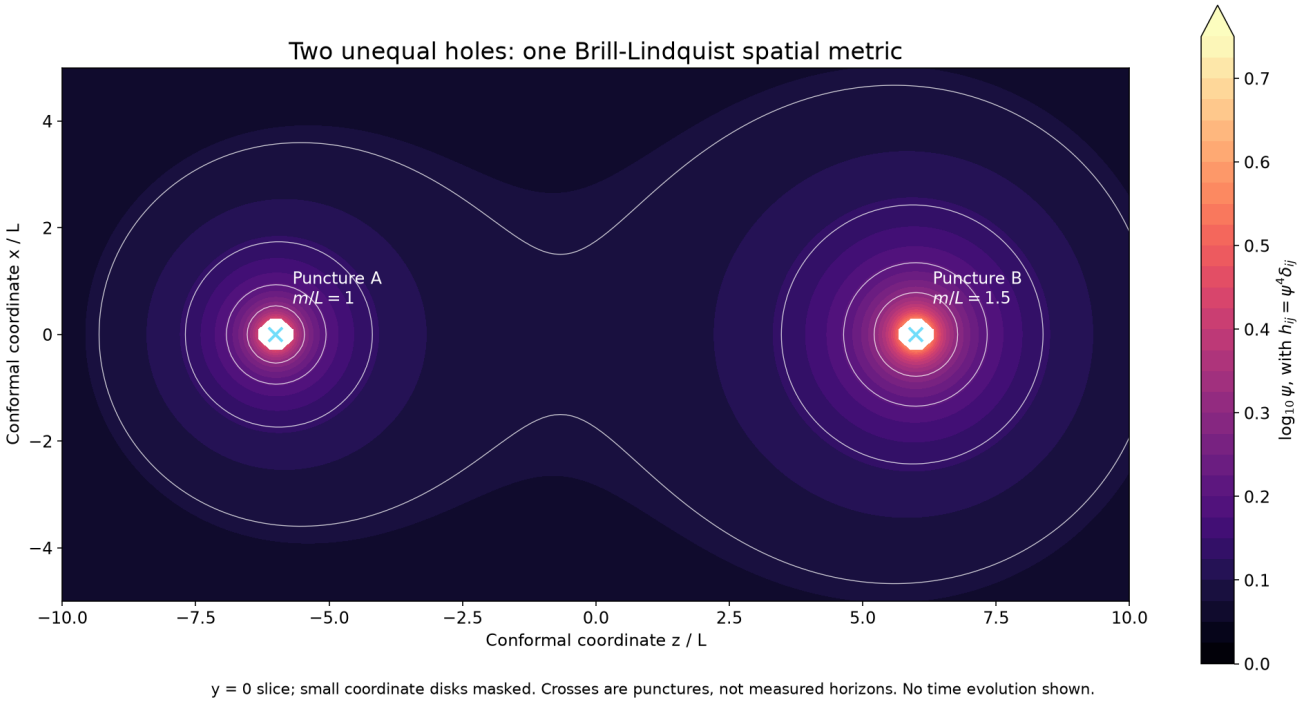


Figure 2. The explicit two-hole initial-data benchmark. This is a nonspherical common parent with unequal ends, not a solved evolution that creates children. Local spherical models must earn an attachment to it or belong to a separate parent family.

The alternative parent route imports an existence result instead of re-solving its gluing equations. Chruściel, Corvino and Isenberg (0909.1101v1) provide sufficiently separated scalar-flat bodies with retained interior regions and a common Schwarzschild outer end. Their more general vacuum-end theorem (1004.1355v1) allows appropriate non-time-symmetric ends. These establish separate families, not a modification that leaves the displayed Brill–Lindquist data unchanged. No explicit sufficient separation is certified here.

8. A failed attachment can sharpen the map

Evidence: radiation junction, moving-boundary conservation/control, and source assumptions.

The first attachment screen pairs a closed radiation FLRW region with a vacuum exterior and a pure constant-tension wall. For a comoving boundary, $a=A\sin\eta$, $t=A(1-\cos\eta)$, and $R=a\sin\chi$. The junction condition would require, over an interval of changing R,

$$k^2R^4 - 2\cos\chi\, kR^3 + 2MR - A^2\sin^4\chi = 0.$$

For nonzero $A\sin\chi$, this cannot be a polynomial identity on that interval. The recorded $\chi=\pi/3$, $M=0.4A$ test exposes a source incompatibility, not a failure of all interiors.

Could a moving wall repair the same model? Intrinsic conservation answers before a new simulation is needed. Fixed pure tension has zero intrinsic stress divergence, so the Codazzi relation requires zero net normal energy flux. In the fluid rest frame,

$$T(U,n)=(\rho+p)\gamma^2 v.$$

A vacuum exterior contributes no flux and radiation has $\rho+p>0$; therefore $v=0$. Motion alone cannot rescue these assumptions. This is a compatibility application of an established conservation law. Allowing a different surface source, exterior state or flux would define a different candidate requiring its own test.

9. A distinct fixed-source wall that does connect

For the next branch, take de Sitter inside and Schwarzschild outside,

$$f_- = 1 - H^2R^2$$
$$f_+ = 1 - \frac{2M}{R}.$$

In $G=c=1$ units let $\kappa=4\pi\sigma>0$, with surface pressure $P=-\sigma$. The signed angular extrinsic curvatures obey

$$\beta_- - \beta_+ = \kappa R$$
$$\beta_\pm = \varepsilon_\pm \sqrt{\dot{R}^2 + f_\pm}\,.$$

Here the dot is wall proper time and $\varepsilon_\pm$ fixes the normal orientation. Squaring and then forgetting those signs would lose physical branches. The useful explicit expressions are

$$\beta_- = \frac{M}{\kappa R^2} + \frac{\kappa^2 - H^2}{2\kappa} R$$

$$\beta_+ = \frac{M}{\kappa R^2} - \frac{\kappa^2 + H^2}{2\kappa} R$$
$$\dot{R}^2 = \beta_-^2 - f_-\,.$$

We hold $H\ell=1$, $\kappa\ell=2$, $M/\ell=0.4$ fixed. On the tested expanding segment $1.2\le R/\ell\le 1.8$, $\beta_->0$ and $\beta_+<0$. The wall proper duration is 0.3718706103ℓ. Direct unsquared and pressure checks, including a separate Christoffel-based calculation, agree within the recorded numerical residuals. The wall's areal-radius rate is not a local three-speed. This published wall family is classified more fully in Section 11; the source has not been retuned to make the segment work.

10. A sourced initial slice and a conditional common parent

Evidence: slice constraints, parent assembly, slice derivation, and end-only gluing argument.

At $R_w=1.5\ell$, the interior is a flat expanding de Sitter cap: a regular spatial three-ball with $h=dl^2+R^2d\Omega^2$, $R'=1$ and $K^i{}_j=-H\delta^i{}_j$. We use $K=-L_T\,h/2$, where T is the future unit slice normal. Define $V=T(R)$. The cap has $V_-=HR$. Its slice determines the wall boost,

$$\gamma = \frac{\beta_- - HR\dot{R}}{1-H^2R^2}$$
$$\gamma v = \frac{\dot{R} - HR\beta_-}{1-H^2R^2}.$$

The exterior inherits

$$R'_{+} = \gamma\beta_{+} - \gamma v\dot{R}$$
$$V_{+} = \gamma\dot{R} - \gamma v\beta_{+}.$$

At this event $\gamma=1.00926203$, $v=0.13516599$, $R'_{+}=-2.02778609$, and $V_{+}=1.90925372$. The exterior slice is therefore not time symmetric at the material wall.

A fixed smooth slicing function tapers $V(R)$ to zero at $R_0=\ell$, before the Schwarzschild throat $2M=0.8\ell$. In the transition,

$$R' = -\sqrt{f_{+} + V^2}$$
$$K^{\theta}{}_{\theta} = -\frac{V}{R}$$
$$K^{l}{}_{l} = -\frac{dV}{dR}.$$

These data describe a slice of the same vacuum Schwarzschild spacetime. They do not interpolate the matter source. Its mass identity is

$$M = \frac{R}{2}(1 + V^2 - R'^2).$$

Where $V=K=0$, the slice continues smoothly through a bridge using $R=2M\cosh^2 u$, with the cap on the negative-u side and the selected parent end on the positive-u side.

The distributional shell constraints are equally necessary. The surface energy and radial momentum on this slice are $E_{\Sigma}=\sigma\gamma$, $J_{\Sigma}=\sigma\gamma\, v$, giving

$$[R'] = -4\pi R E_{\Sigma}$$
$$[V] = 4\pi R J_{\Sigma}.$$

Their recorded residuals are below 2×10^{-15} in normalized units; independent finite-difference bulk residuals are below 10^{-10}. These are implementation checks, not a global stability estimate.

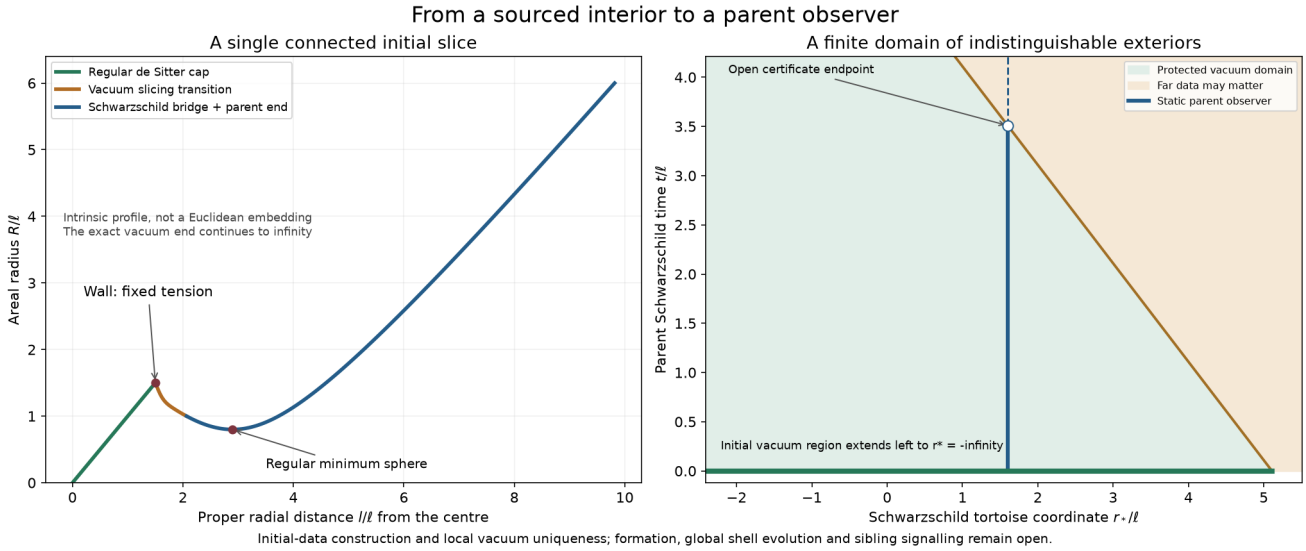


Figure 3. The verified initial connection. Spatial distances are not ages. Because |R'|>1 on part of the chosen slice, the radius profile should not be interpreted as a Euclidean surface of revolution. The protected right-exterior observer interval is a separate causal-domain statement.

The selected right end of this body *A*, of mass *0.4ℓ*, can be used with the selected right end of an ordinary complete Schwarzschild bridge *B*, of mass *0.6ℓ*, in the imported vacuum-end gluing theorem. It acts on smooth vacuum ends and preserves an open overlap. Reattaching unchanged interiors on identical full *(h,K)* data leaves the original shell delta source intact and creates no seam source. We do not apply a smooth global-data corollary directly to a thin shell.

The resulting conditional initial-data family has one designated common parent end **and the untouched other end of** *B*: two asymptotic ends in total. That second end has not become a cosmological child. Sufficiently large separation is required but no numerical threshold or explicit multi-body development has been calculated. The achievement is a compatible sourced initial connection, not a regular formation history, a general Einstein–shell evolution theorem or a stable cosmological future.

IV. What the observers can reach

11. Complete one reference before assigning its causal arrows

Evidence: isolated causal completion, observer access, and protected parent interval.

The wall's isolated spherical completion is the flat-exterior limit of solution 13 in Aguirre and Johnson, gr-qc/0508093v2, Figures 7 and 10. Writing $x=R/\ell$, its expanding branch has

$$\dot{R}^2 = \frac{25x^2}{16} - 1 + \frac{3}{10x} + \frac{1}{25x^4} > 0.$$

It starts at a finite proper-past *R=0* boundary in the Schwarzschild white-hole sector, crosses *R=2M* into the left exterior, and crosses the de Sitter horizon into an expanding interior chart. It reaches arbitrarily large radius at infinite wall proper time. Its Schwarzschild curvature $48M^2/R^6$ diverges at the past boundary. The regular cap on the chosen slice therefore does not imply a regular past formation. The retained reference also contains the usual future Schwarzschild black-hole singular boundary.

Choose three clocks. *C* is the cap center, with de Sitter flat proper time t_C. *W* carries wall proper time τ_W, set to zero at $R=1.5\ell$. *P* is static in the right exterior at $R_P=1.6\ell$, with proper time $\tau_P=\sqrt{1-2M/R_P}\ t_{Schw}$. Their origins coincide with a selected slice convention; this is not an equality of universe ages.

For the flat de Sitter chart,

$$\eta = -\ell e^{-t_C/\ell}$$
$$R = ar$$
$$a = e^{t_C/\ell}.$$

Radial null lines have constant η-r or η+r. At W_0, η=-ℓ and r=1.5ℓ. Its inward ray would reach η=0.5ℓ, outside the chart's future η<0; hence it never reaches C. A central signal emitted at t_C=0 likewise cannot catch the future wall.

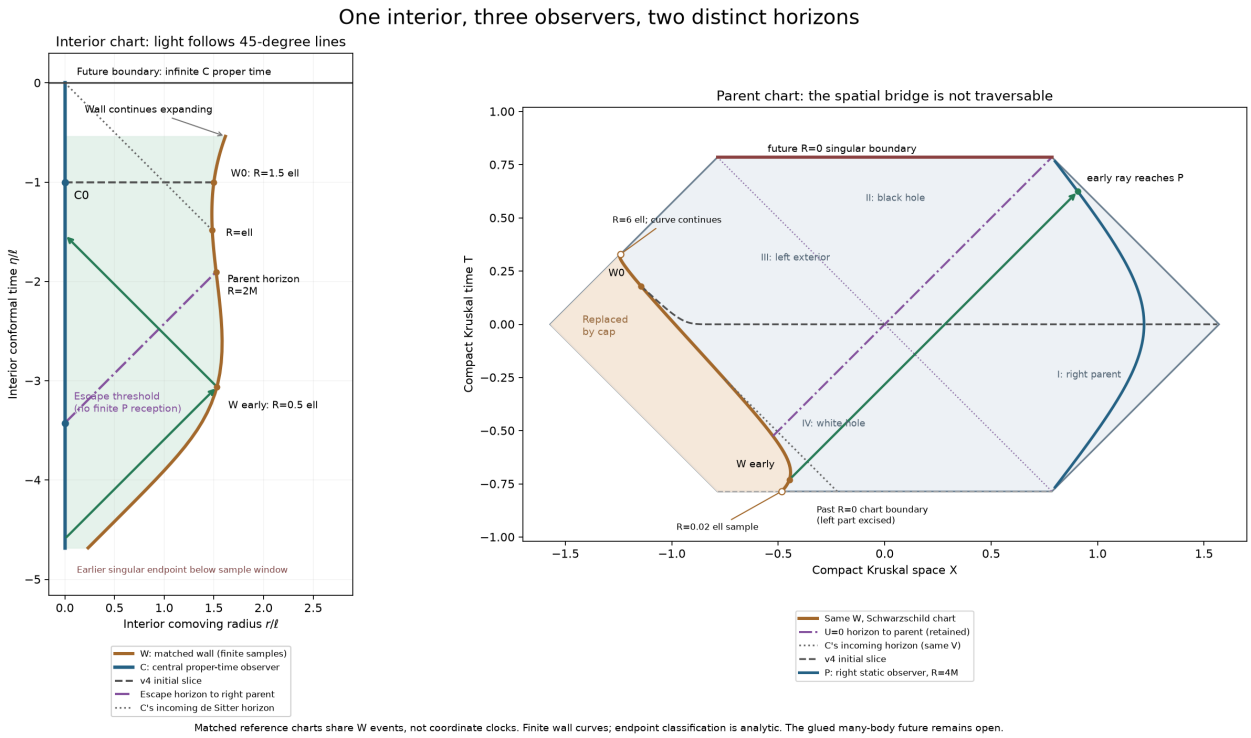


Figure 4. The isolated reference in joined causal charts. Horizons are clipped to retained domains; the central cosmological horizon continues into the Schwarzschild chart along its appropriate constant null coordinate. The faint past R=0 line also shows uncut chart context, part of which is excised. Open wall endpoints at R/ℓ=0.02 and 6 mark finite plot samples, not physical beginnings or endings.

Emission and destination	Result in this reference	Required distinction
New $C\rightarrow W$ or $W\rightarrow C$ emission at or after the anchor slice	No future reception	Earlier signals may arrive later
Right parent $P\rightarrow W$ or $P\rightarrow C$	No future causal route	Left and right exterior regions are different
Early $W\rightarrow C$, while $R<\ell$	Allowed	Reception time depends on the emission event
Early $C\rightarrow W\rightarrow P$, crossing while $R<2M$	Allowed	The limiting horizon ray has no finite parent reception
These same arrows in a glued multi-body future	Unresolved	Initial-data gluing does not supply that global future

For a wall event with interior time t_W, the central emission reaching it and the reception of its inward ray satisfy

$$t_e = t_W - \ell \ln(1 + R/\ell)$$
$$t_r = t_W - \ell \ln(1 - R/\ell).$$

The second relation requires $R<\ell$. At the early event $R=0.5\ell$, the recorded central emission is -1.5242168241ℓ and inward reception is -0.4256045355ℓ. An outward early route through that event reaches P at its own proper time $+2.5311656510\ell$. Thus exclusion of **new emissions** after a slice must never be rewritten as exclusion of all later receptions. Emissions approaching the central horizon from below can arrive arbitrarily late.

The critical central emission for escape to the right parent is $t_C=-1.2312944525\ell$. Equality follows the parent horizon and has no finite parent reception. That horizon extends into the cap along $\eta-r=\eta_W(2M)-r_W(2M)$; it is not the surface $R=2M$ everywhere. The central observer's de Sitter horizon $\eta+r=0$ is a different boundary. Detailed clock domains and finite checks are retained in the observer note.

12. Different exteriors can be locally indistinguishable

There is also a result that does not require the glued parent's full future. Retain the exact right Schwarzschild $K=0$ slice for $2M<R<R_{cut}$, with $R_{cut}=10M$. For a static observer at $R_{obs}=4M$, the exterior cutoff cannot influence it within

$$0 < t_{\mathrm{Schw}} < r_*(R_{\mathrm{cut}}) - r_*(R_{\mathrm{obs}})$$
$$r_*(R) = R + 2M \ln(R/(2M) - 1).$$

The observer's proper duration is the right side multiplied by $\sqrt{1-2M/R_{obs}}$. For $M=0.4\ell$, the bound is $t_{Schw}<3.5090354889\ell$, or $\tau_P<2.4812627896\ell$. The inner optical boundary is at negative infinity, so it introduces no finite inner cutoff.

Shared smooth vacuum data give the same vacuum development in this domain. The owning argument uses standard Cauchy-development uniqueness, with Sbierski (1309.7591v2) as a source. The bound ends a guaranteed interval; it does not predict that a signal actually arrives then. Matching additional fields and their states requires additional data.

This supplies one precise atlas of indistinguishable exteriors. Distinct larger assemblies can share a protected local geometry. The next useful question is which later measurement or extra state datum separates them. Failure to distinguish exteriors can be a result, rather than a reason to add unconstrained parameters until a preferred story appears.

V. Fields, transfer and finite memory

13. A first field that reaches the interior

Evidence: field-transfer record and field-transfer owner.

On the fixed de Sitter/Schwarzschild wall, retain $H\ell=1$, $k\ell=2$, $M/\ell=0.4$ and a real classical massless scalar in its spherical mode, with conformal coupling $\xi=1/6$ and no additional scalar wall interaction. For $\chi=R\varphi$, the distributional scalar equation gives

$$[\phi] = 0$$
$$[n\phi] = 6\xi k\phi$$
$$[\chi] = 0$$
$$[n\chi] = (6\xi - 1)k\chi = 0.$$

The conformal interface barrier cancels; the smooth exterior potential $f_+ 2M/R^3$ remains. A regular center has $\chi=F(\eta+r)-F(\eta-r)$. For incoming wall characteristic $I=(d/d\tau_W+n)\chi$, its exact conditional response is

$$\phi_C[T(\tau_W)] = \frac{I(\tau_W)}{T'(\tau_W)}$$
$$T = t_W - \ell \log(1 - R/\ell)$$
$$T' = \frac{\gamma(1+v_{\rm pec})}{1-R/\ell} > 0.$$

The prescribed wall-clock calibration has half-width 0.03ℓ, duration stretch 2.63230219 and normalized extrema $+23.84144800$ and -24.46533478. Its central field vanishes after the received support through $t_C=0$, conditional on zero incoming data afterward and the proved no-return interval. These are field amplitudes, not energy fractions. The calibration does not generate its exterior input.

13.1. Where the early signal can originate

Evidence: preparation record and preparation owner.

Independent data can instead be placed in retained white-hole sector IV. The ordinary right-parent observer remains causally excluded. Let $x=R/\ell$, $F(x)=(1-x/0.8)e^{x/0.8}=UV$, $V_a=V_W(0.45)$, $V_b=V_W(0.55)$ and $U_0=F(0.4)/V_a$. Define D by $V_a\le V\le V_b$ and $U_0\le U\le U_W(V)$; there $0.4\le R/\ell\le 0.55$. Prescribe zero bottom and cap data and $g(V)=\varepsilon\, B[4(V-V_m)/(V_b-V_a)]$ on $U=U_0$, where $B(z)=(1-z^2)^4$ for $|z|<1$ and zero otherwise. This is a compact C^3 input, independent of any desired central waveform.

With $L(u)=V_a$ below U_a and the inverse wall graph above it,

$$\chi = g - \int_{U_0}^{U} du \int_{L(u)}^{V} dv\, K\chi$$
$$K = \frac{16M^4}{R^4} e^{-R/(2M)}$$
$$\chi_U|_W = 0.$$

The integral operator has analytic uniform-norm bound

$$q_* = 16e^{-1/2}[\sqrt{F(0.4)} - \sqrt{F(0.55)}]^2 = 0.13882099234357653 < 1.$$

It therefore determines a unique finite transfer. The central arrival window is $-0.581758951595344\le T/\ell\le -0.250230700560700$; the no-return null-label gap is 2.634591912177392ℓ. Remote physical generation and data beyond that window remain unspecified.

13.2. A prepared pulse leaves a finite scattering tail

Evidence: prepared-response record and response owner.

For this unchanged input, let A denote the positive integral operator above. After the compact input ends, its Neumann series gives $-Ag\le\chi\le-(1-q_*)Ag<0$. Hence the determined wall derivative

$$J = g'(V) - \int_{U_0}^{U_W(V)} K\chi\, dU > 0$$
$$(1-q_*)\mathcal{B}(V) \le J(V) \le \mathcal{B}(V), \quad \mathcal{B} = \int KAg\, dU$$

produces a positive central tail on $-0.340772594886934<T/\ell\le -0.250230700560700$. The sampled normalized extrema are 47.29833694 and -45.69595941; the final field is 0.00129049806. At the final

point, $J/\varepsilon=0.0008020390045$ lies in the independently evaluated bracket $[0.0006965878198,0.0008088769160]$. The 257-to-513 spatial and 129-to-257 geometry comparisons change the central field by at most 2.79286258×10^{-7} and 3.40622250×10^{-8}. The calculation uses 5,597,264 kernel values within its ten-million/60-CPU-second limits. These numerical controls support the finite waveform; the sign follows from the analytic argument. No permanent memory, continuum enclosure or later decay law is established.

13.3. What the signal determines when preparation is unknown

Evidence: inverse record and inverse owner.

In this fixed geometry, put $f=\chi/\varepsilon$, $h=f(U_W(V),V)$, $\theta=T/\ell$ and $Y=\ell\varphi_C/\varepsilon$. The zero cap fixes the integration constant:

$$h(V)=\tfrac{1}{2}\int_{\theta_a}^{\theta(V)}Y(s)ds$$
$$f=(I-\mathrm{B})^{-1}Eh$$
$$g(V)=f(U_0,V)$$

$$(\mathrm{B}f)(U,V)=\int_U^{U_W(V)}dp\int_{L(p)}^{V}dq\,K(p,q)f(p,q).$$

The operator-norm bound $q_*<1$ gives a bijection between initially quiet C^2 source traces and compatible C^1 signed central histories. For nonzero nonnegative compact sources, the post-input restrictions are $h<0$, $Y>0$ and $\int_{\theta_a}\hat{}(\theta(V))Yds=2h<0$. An arbitrary fitted trace need not satisfy them. This analytic source inverse neither identifies geometry nor reconstructs a source numerically from sampled or intensity-only data.

13.4. The transmitted probe and the wall's stress

Evidence: stress record and wall-stress owner.

Fix canonical kinetic normalization, $\kappa_4=8\pi G$ and $f(\varphi)=1/(2\kappa_4)-\varphi^2/12>0$ in the weak-field interpretation. The full stress includes the nonminimal Einstein term once:

$$T^{\phi}_{\mu\nu}=\nabla_\mu\phi\nabla_\nu\phi-\tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2+\xi[G_{\mu\nu}\phi^2+(g_{\mu\nu}\Box-\nabla_\mu\nabla_\nu)\phi^2].$$

For the original sufficiently regular traces, $d=[n\varphi]=k\varphi$ and $C_{ab}=[K_{ab}]-\gamma_{ab}[K]=2k\gamma_{ab}$ give $S^{\varphi}{}_{ab}=\xi(2\varphi d\gamma_{ab}-\varphi^2C_{ab})=0$ and $[T^{\varphi}{}_{na}]=(1-6\xi)k\varphi\, D_a\varphi=0$. Thus no additional scalar surface source or tangential flux jump is required at quadratic order. The individual fluxes and smooth bulk gravitational source need not vanish.

13.5. The prepared pulse requires gravitational initial data

Evidence: initialization record and initial-data owner.

On the source leg, keep the reference lapse, radius 0.4ℓ, both corner expansions and affine normalization. Use the affine background radius $x=R_0(U_0,V)/\ell$, $V(x)=F(x)/U_0$ and $b(x)=B(4[V(x)-V_m]/(V_b-V_a))$; primes here and in section 13.6 mean d/dx. With $p=b(x)/x$, $\varphi=(\varepsilon/\ell)p+O(\varepsilon^3)$ and $\alpha=\kappa_4\varepsilon^2/\ell^2$, write $R/\ell=x+\alpha\, h+O(\alpha^2)$. The intrinsic constraint forces

$$h'' = -\tfrac{x}{3}p'^2 + \tfrac{x}{6}pp''$$
$$h(0.4) = h'(0.4) = 0.$$

After the compact trace, $h'<0$ and $h<0$ by the integrated expressions; h' becomes constant. At $x_b=0.4815042082576364$, $h=-6.12283354819419$ and $h'=-162.18055362491776$. Direct integration, integration by parts and the transformed affine Einstein-frame equation agree. These are initial radius coefficients, not a mass shift or an evolved wall; no finite amplitude or perturbative remainder bound is selected.

13.6. An invariant mass response on the same initial surface

Evidence: source-mass record and mass owner.

With $m=M/\ell=0.4$, the fixed transverse corner datum and scalar equation supply $d=(a/x-1)/4$, $a=0.8$, and $s'=-(s+dp')/x$, $s(0.4)=0$. For the geometric mass length, use

$$\mathfrak{m} = \tfrac{R}{2}[1 - g^{ab}\partial_a R \partial_b R]$$
$$\tfrac{\mathfrak{m}_J}{\ell} = m + \alpha\mu_J + O(\alpha^2).$$

The borrowed flux and full conformal frame conversion yield

$$\mu_E = -\tfrac{1}{4}\int_{0.4}^{x} t(a-t)p'(t)^2 dt$$
$$\mu_J = \mu_E + \tfrac{mp^2}{12} + \tfrac{x^2 p}{3}(s + dp').$$

After the input, the frame masses agree at the constant negative coefficient $\mu_J=\mu_E=-28.88319705406449$. Their largest sampled difference during input is 4.265459834842238. Independent mixed-radius and improved stress laws agree. This is a finite source-leg response relative to the reference mass, not a negative total mass, vacuum future or asymptotic energy.

13.7. The scalar adds no normal-force jump

Evidence: normal-force record and normal-force owner.

The Gauss jump $[G_{nn}]=-2k\bar{K}$ and complete scalar stress give

$$[T^{\phi}_{nn}] = 2\xi(6\xi - 1)k\bar{K}\bar{\phi}^2 = 0$$
$$S^{ab}\bar{K}_{ab} = [T_{nn}].$$

The cap contribution $G^{-}_{nn}=-3H^2$ is essential. Fixed vacuum and material tension therefore require $\delta\bar{K}=0$ at order ε^2. This mean-curvature variation includes the bulk metric, induced metric and wall displacement. It does not require zero displacement or an unchanged trajectory.

13.8. Exterior mass closes from the known wall field

Evidence: analytic closure record and wall-mass owner.

On the smooth exterior wall limit in the same $0.45\le R/\ell\le 0.55$ window, put $s=\tau/\ell$, $x=R/\ell$, $v=dx/ds$, $\beta=\beta_+$, $w=\chi_W/\varepsilon$, $j=dw/ds$ and $p=w/x$. The two known derivatives are $A=(j-vp)/x$ and $B=(j-\beta p)/x$. Both null mass fluxes and the complete conformal endpoint term give

$$L = \frac{x^2}{2}\left[\beta AB - \frac{v}{2}(A^2 + B^2)\right]$$
$$Q = \frac{mp^2}{12} + \frac{x^2 p}{6}(vA - \beta B)$$
$$\mu_J(s) = \int_{s_a}^{s} L(s')ds' + Q(s), \quad Q(s_a) = 0.$$

The original-frame improved stress independently gives $d\mu_J/ds=L+dQ/ds$. Neither expression divides by β. This analytic conditional functional is evaluated with the coupled quadratic geometry in Sections 13.10--13.12. The wall tail alone supplies no late mass sign or plateau. Physical preparation selection, the next scalar amplitude, stability and a finite-amplitude validity bound remain open. Detailed scalar-chain derivations belong to their owners; historical supporting calculations are cited at Zenodo.

13.9. Cap mass and a regular wall displacement functional

Evidence: cap-wall record, model specification snapshot and cap-wall owner. The common model interface separates the fixed action, prescribed preparation and unit scale from derived responses. Its parameters $h=H\ell=1$, $b=k\ell=2$ and $m=M/\ell=2/5$ remain declared inputs. We retain the same canonical real massless scalar, conformal coupling, scalar-independent vacuum terms and constant material tension in the Jordan frame.

Let $r=x+\alpha\rho+O(\alpha^2)$ be the actual wall radius in units of ℓ, with $\alpha=\kappa_4\varepsilon^2/\ell^2$ and proper wall time $s=\tau/\ell$. The two geometric masses have expansions

$$\frac{\mathfrak{m}_+}{\ell} = m + \alpha\mu_+ + O(\alpha^2)$$
$$\frac{\mathfrak{m}_-}{\ell} = \frac{h^2 r^3}{2} + \alpha\nu_- + O(\alpha^2).$$

Here μ_+ is the exterior coefficient μ_J of Section 13.8. The cap vacuum is evaluated at the actual radius: its total quadratic mass coefficient on the reference wall is $\nu_-+3h^2x^2\rho/2$. Holding that vacuum mass constant would omit a term required by the angular junction. The shared quiet corner fixes $\mu_+(s_a)=\nu_-(s_a)=\rho(s_a)=0$.

Using the signed normals $\beta_-=\beta_+ +bx$, define $B_-=(j-\beta_- p)/x$ with the same $p=w/x$, $j=w_s$ and $A=p_s$ as above. The cap's vacuum-subtracted mass closes as

$$L_- = \frac{x^2}{2}\left[\beta_- AB_- - \frac{v}{2}(A^2 + B_-^2)\right] + \frac{h^2x^2vp^2}{2}$$

$$Q_- = \frac{x^2 p}{6}(vA - \beta_- B_-) - \frac{h^2x^3p^2}{12}$$
$$\nu_-(s) = \int_{s_a}^{s} L_-(s')\,ds' + Q_-(s).$$

Direct differentiation reproduces the Jordan-frame improved-stress law $\nu_{-,s}=x^2(\beta_- q_- -vP_-)/2$. Here $q_-=t_{uN,-}$ and $P_-=t_{NN,-}$ are the flux and normal pressure of $t_{ab}=\ell^4 T^{\varphi}{}_{ab}/\varepsilon^2$. The extra vacuum terms in L_- and Q_- are essential; the exterior's zero-vacuum shortcut cannot be carried into the cap. The evaluation uses first scalar derivatives, while covariant Hessians enter the independent transport check. No derivative of sampled waveform data is used.

For this conformal scalar, $f+3f'^2=1/(2\kappa_4)$ reduces the scalar-tensor pure-tension junction to the same signed angular law. With $D=\mu_+ -\nu_-$, its linearization gives

$$\delta\beta_{\pm} = \beta_{\pm,x}\rho + \frac{D}{bx^2}$$
$$\rho_s = v_x\rho + \frac{\mu_+/x + \beta_+ D/(bx^2)}{v}.$$

The cap expression replaces the numerator by $v_-/x+\beta_- D/(bx^2)$ and agrees exactly. Because $v=x_s>0$ on $x\in[0.45,0.55]$, the equation stays regular through $\beta_+=0$. Its homogeneous solution is $C\ v(s)$, a relative proper-time origin shift removed by the shared quiet event.

The existing scalar junction identities give equal one-sided fluxes and pressures, $q_+=q_-=q$ and $P_+=P_-=P$. Subtracting the two mass laws yields $D_s=-bx^3q/2$. Together with the displacement equation, this gives equal corrections to the dimensionless normal accelerations $a_\pm$ and zero change in the dimensionless mean extrinsic-curvature trace:

$$\delta a_+ = \delta a_- = \frac{6m\rho}{bx^4} - \frac{2D}{bx^3}$$
$$\delta\overline{K} = 0.$$

Thus the temporal jump and normal-force condition close with the angular condition and mass transport, without setting $\rho=0$ or adding a freely chosen cap mass or wall force. These are formal quadratic identities, checked by independent algebra, not an evaluated displacement curve or a completed sourced spacetime.

The next connection is the smooth bulk metric on both sides; the exterior characteristic calculation below supplies its first evolved domain. Geometric mass alone does not determine the null lapse, the wall's null-coordinate embedding or the central observer's clock. Their reconstruction must use the shared wall data and characteristic initial data before calculating the corrected arrival map and subsequent scalar response. A nonzero allowed amplitude, a controlled remainder, formation and stability remain separate questions.

13.10. The prepared field drives a smooth exterior metric response

Evidence: metric refinement, field refinement, derivative-quadrature comparison and bulk-response owner, with separate geometry, source quadrature and time tolerance comparisons.

The gravitational data of Sections 13.5--13.6 determine a computed smooth exterior response at quadratic order on $[U_0,U_a]\times[V_a,V_b]$, whose two past null sides are specified. This rectangle lies inside the retained wedge. Its final transverse leg $U=U_a$ supplies data for a later extension toward the moving wall; no exterior metric is continued through the excised cap.

Keep the original null coordinates and fixed Jordan source lapse. With $r_0=R_0/\ell$, write $r_E=r_0+\alpha\ H$ and $log\ C_E=log\ C_0+\alpha\ c$ to the retained order. The leading scalar is $\varphi=(\varepsilon/\ell)p$, with both first derivatives obtained from its characteristic integral equations. The existing smooth exterior Einstein equations simplify under $y=r_0H$, $z=c+H/r_0$ to

$$y_{UV} = \frac{C_0}{4r_0^2}y - \frac{C_0}{4}z$$
$$z_{UV} = -\frac{3C_0}{4r_0^4}y + \frac{C_0}{4r_0^2}z - p_Up_V.$$

Integrating their right sides in U gives a linear evolution in V. The quiet bottom has $y=z=0$. At $U=U_0$, let $x=r_0(U_0,V)$ and let h_{init} denote the original Jordan radius coefficient from Section 13.5. The source values in the unchanged coordinate gauge are

$$y_0 = x h_{\text{init}} - \frac{x^2 p^2}{12}$$
$$z_0 = \frac{h_{\text{init}}}{x} - \frac{p^2}{4}.$$

The frame and affine-parameter corrections must both be retained; the Einstein-affine radius coefficient alone is not the source value at fixed original coordinates. Returning to the Jordan metric gives $h_J=H+r_0p^2/12$ and $c_J=c+p^2/6$, where $r_J=r_0+\alpha\ h_J$ and $log\ C_J=log\ C_0+\alpha\ c_J$.

At (U_a,V_b), the 65-node metric calculation with 257 field nodes reports the following dimensionless quadratic coefficients:

Quantity	Recorded coefficient
Radius, h_J	*-5.7131513827*
Logarithmic null lapse, c_J	*-1.5461215575*
Geometric mass, μ_J	*-28.8833375991*

Changing the metric grid from 33 to 65 nodes reduces the largest sampled mass-transport residual from 1.65×10^{-6} to 4.31×10^{-7}. On the common sampled grid, the changes in h_J,c_J,μ_J are respectively at most 1.15×10^{-7}, 3.82×10^{-6} and 1.23×10^{-6}. A separate 129/257 field-grid comparison changes these coefficients by at most 2.14×10^{-8}, 9.55×10^{-8} and 5.78×10^{-7}; the 32/64 derivative-quadrature comparison changes them by less than 7.2×10^{-15}. Both null constraints are checked in integral form, and geometric mass is compared with transport along both null directions. The radial lapse reconstruction checks the same U constraint in different variables.

Changing geometry interpolation and source quadrature changes these three coefficients by less than 7.2×10^{-15} and 1.5×10^{-13} respectively; tightening the time tolerance tenfold changes them by less than 1.21×10^{-11}. Each complete two-solve comparison took between 17.16 and 21.91 process-CPU seconds within its 120-second budget.

These are finite numerical comparisons, not rigorous error enclosures or a selected nonzero physical amplitude. They establish a sourced exterior metric response on the rectangle and provide its transverse radius/lapse data. Sections 13.11--13.12 continue to the moving wall and cap using the same source and material junction. The full finite-amplitude sourced configuration, corrected observer signal and finite-amplitude remainder have not yet been established.

13.11. The same field displaces the wall and changes its exterior clock

Evidence: field refinement, radial quadrature, clock integration and the wall-evolution owner.

The leading field now supplies a computed response on the moving boundary, with the action, source and quiet corner of Sections 13.5--13.10 unchanged. Extending its first-derivative evaluator to the retained wedge requires the correct lower endpoint: a column with $U>U_a$ begins where it meets the wall. At that event $\chi_U=0$, while $p_U=-p\ r_{0,U}/x$ is generally nonzero. Both derivatives are obtained from the integral equations of the original field solution.

Define the areal-null lapse $Z=C/(2r_U)$. The Einstein-frame mass and logarithmic lapse coefficients follow radial constraints at fixed V:

$$\mu_E = \mu_{E,0} - \int_{U_0}^{U} \frac{r_0^2 r_{0,V}}{C_0} p_U^2\, dq$$
$$\zeta_E = \zeta_{E,0} + \int_{U_0}^{U} \frac{r_0}{2r_{0,U}} p_U^2\, dq.$$

Their source constants are already fixed by the characteristic data. At the wall, frame conversion gives $\mu_+^{radial}=\mu_{E,W}+Q_+$ and $\zeta_{J,W}=\zeta_{E,W}+p^2/4$. The independent timelike mass functionals of Section 13.9 determine μ_+ and ν_-, and hence the quiet-anchored displacement ρ. The radial and timelike mass values are compared, rather than equated by the numerical implementation.

The exterior embedding also follows from these data. Write $V_W=V_0(1+\alpha\eta)$, with s the original Jordan wall proper time, $d=v-\beta_+>0$, and $Z_0=-2a/V_0$, $a=2m$. Expanding $V_{W,s}=1/[Z_J(r_{W,s}-\beta_+)]$ gives

$$\eta_s = \frac{1}{2ad}\left[\zeta_{J,W} + \frac{\rho_s - \delta\beta_+}{d}\right]$$
$$\eta(s_a) = 0.$$

The background lapse is evaluated at the moved V_W in this expansion. The equation stays regular through the zero of the signed β_+.

At the final reference radius $x=0.55$, the 257-node field calculation gives:

Dimensionless quadratic coefficient	Recorded value
Exterior mass, μ_+	-28.8834223256
Cap excess mass, ν_-	17.7590784205
Wall displacement, ρ	-5.2868390417
Exterior clock ratio, $\eta=V_1/V_0$	-0.8178125677

For example, the radius expansion is $r_W=0.55+\alpha\rho+O(\alpha^2)$; the tabulated number is its coefficient, with no finite α selected. The cap excess subtracts its vacuum mass at the actual wall radius.

Changing the leading field from 129 to 257 nodes reduces the largest sampled mass-path discrepancy from 6.49×10^{-7} to 1.63×10^{-7}. The maximum changes at nine common time cuts in μ_+,ν_-,ρ,η are respectively 3.38×10^{-7}, 2.09×10^{-7}, 3.64×10^{-8} and 3.57×10^{-9}. Doubling the clock panels from 8 to 16 changes η by at most 2.41×10^{-8}. The radial 2/4-point comparison changes it by less than 1.35×10^{-13}. The angular jump, two forms of the displacement equation and clock normalization are checked on the same trajectories.

The 32/64 field-derivative comparison leaves the eight reported comparison quantities identical in this run. Separate geometry, source and ODE tolerance comparisons change them by less than 5.64×10^{-11}, 2.85×10^{-14} and 8.25×10^{-13} respectively. All seven comparisons completed within their individual 120 process-CPU-second budgets. An independent rectangle overlap finds mass and lapse differences of at most 4.08×10^{-7} and 1.30×10^{-6} against the mixed metric solver. Different radial quadratures expose error that a shared-rule constraint residual can cancel; that residual alone is not a total metric error estimate.

These computations connect the sourced exterior and the two-sided mass functionals to an actual wall displacement and exterior null embedding at quadratic order. Their numerical differences are refinement diagnostics. Section 13.12 supplies the cap metric and central arrival correction. The next scalar waveform and finite-amplitude remainder remain to be calculated.

13.12. The cap geometry changes the central arrival clock

Evidence: field refinement, radial integration, further radial refinement, ray integration, further ray refinement, wall-time integration, geometry, derivative quadrature, centre quadrature, centre threshold and the cap derivation.

The same prepared wall trace fixes the leading regular field in the cap. With $A_0=1-r^2$, $r_*=\operatorname{atanh} r$ and $f(q(V))=\chi_W(V)/\varepsilon$, where q is the existing central arrival map,

$$p(T,r)=\frac{f(T+r_*)-f(T-r_*)}{r}$$
$$p(T,0)=2f'(T).$$

The reflected term remains in the interior. Both profile derivatives and stable centre evaluations come from the existing field's continuum integral equations. The fixed cap vacuum transforms to a quadratic potential $H^2\varphi^2$ in the Einstein frame; it is retained in the metric.

For the polar-areal metric $ds_E^2/\ell^2=-e^{2\alpha\delta_E}A\,dT^2+A^{-1}dr^2+r^2d\Omega^2$, write $A=A_0-2\alpha\nu_E/r$. The radial constraints give

$$\nu_{E,r}=\frac{r^2}{4}\left(\frac{p_T^2}{A_0}+A_0p_r^2\right)+\frac{r^2p^2}{2}$$
$$\delta_{E,r}=\frac{r}{2}\left(p_r^2+\frac{p_T^2}{A_0^2}\right).$$

Regularity fixes $\nu_E(T,0)=0$. Taking T to be Jordan central proper time fixes $\delta_E(T,0)=-p(T,0)^2/12$. These constants require no new source history. At the wall the Jordan excess is $\nu_J=\nu_E-r^2A_0pp_r/6-r^3p^2/12$; its endpoint term agrees with Q_-, while the independent radial ν_E agrees with $\int L_-ds$. The scalar equation gives the corresponding mass-conservation identity.

Let the cap wall time be $T_W=T_0+\alpha\,T_1$ and its Einstein radius be $x+\alpha\,e$, where $e=\rho-xp_W^2/12$. The material normalization fixes T_1 from the same mass, lapse and wall displacement. The inward null equation then gives the central arrival coefficient at fixed wall proper time:

$$\Delta_C=T_1+\frac{e}{A_0(x)}+\int_0^x\left[\frac{2\nu_E}{rA_0^2}-\frac{\delta_E}{A_0}\right]_{T=q_0-r_*(r)}dr$$
$$q_0=T_0+r_*(x).$$

The integral is regular at the centre. At a fixed exterior null label, the moved emission event adds the correction

$$\Delta_C|_V=\Delta_C-q_{0,s}\frac{V_1}{V_{0,s}}$$
$$T_C(V)=q_0(V)+\alpha\Delta_C|_V+O(\alpha^2).$$

At the final reference wall event, the primary 257-node calculation with 16-point radial/ray rules and 16 cap clock panels gives:

Dimensionless quadratic coefficient	Recorded value
Cap Einstein excess mass, ν_E	*17.7590647726*
Cap lapse, δ_E	*103.4370285720*
Wall cap-time shift, T_1	*-8.4727761614*
Central arrival at fixed wall proper time, Δ_C	*-128.0670033577*
Central arrival at fixed exterior label	*-130.3664151052*

The maximum radial/timelike cap-mass discrepancy is 2.52×10^{-9} over the recorded cuts and cap clock quadrature nodes. Changing the field from 129 to 257 nodes changes the fixed-label arrival coefficient

by at most 6.42×10^{-7}. The clock convention matters: the final exterior-label shift alone contributes 2.2994117476 to the difference between the two arrival coefficients.

The 8/16 ray refinement changes the arrival coefficient by at most 2.31×10^{-5}; the further 16/32 ray comparison with 129 field nodes changes it by 1.84×10^{-9}. At that fixed field resolution, 129/257 geometry knots change it by 1.06×10^{-8}, and moving the stable-centre threshold from 10^{-4} to 10^{-3} changes it by 9.53×10^{-10}. Doubling the centre quadrature changes it by 7.11×10^{-15}; doubling the differentiated-integral quadrature leaves the saved coefficients equal at floating-point precision. The coarser 8/16 radial comparison changes the arrival coefficient by 6.30×10^{-3}; a separate 16/32 radial comparison at 129 field nodes reduces that sensitivity to 1.97×10^{-9}. The 8/16 cap time-panel comparison changes it by 1.92×10^{-5}, exceeding the field and finer radial/ray sensitivities. Its two time integrations reuse identical reference rays, with fresh and reused work recorded separately. These differences do not form a rigorous total error bound.

The metric ray integrand has both mass and lapse terms. With $D=\delta_E-2\nu_E/(rA_0)$, the radial constraints independently give $D_r=2\nu_E(1-3r^2)/(r^2A_0^2)-rp^2/A_0$ and the integrand is $-D/A_0$. Positive cap mass does not fix its sign. The records preserve signed quadrature contributions and sampled extrema; at the final primary cut the ray integral is -112.0145129711.

The comparisons retain explicit fixed settings. Geometry, cap derivative, centre and further-radial/ray controls use 129 field nodes to fit their complete pairs within the declared budget; the main field comparison uses both 129 and 257. Refinement differences are numerical sensitivity estimates. No finite α is selected, and this metric null-arrival correction does not include the order-$\varepsilon\alpha$ scalar amplitude. Its perturbed wave equation and source continuation are specified in the next section. Propagating that correction and establishing finite-amplitude control remain the next physical connection.

13.13. Geometry supplies an outgoing scalar boundary correction

Evidence: field refinement, radial integration, ray integration, further radial refinement, further ray refinement and the scalar-metric derivation.

Keep the same prescribed reduced-field input exactly on the original anchored source and write $\chi/\varepsilon=c_0+\alpha\, c_1+O(\alpha^2)$. Conformal covariance preserves χ between the two frames. In cap null coordinates normalized to the central clock, $c_0=f(q_+)-f(q_-)$ and the computed metric gives

$$\partial_-\partial_+\widetilde{c}_1=-K_1c_0$$
$$K_1=\frac{A_0}{4}\left[\frac{2\nu_E}{r^3}-\frac{p^2}{3}-\frac{p_T^2}{6A_0}+\frac{A_0p_r^2}{6}\right].$$

The retained wall interval has $q_{-,W}<q_a$. Its outgoing homogeneous correction is quiet, so the returning derivative is determined before the incoming correction is solved:

$$I_-=-\int_{q_a}^{q_{+,W}}K_1(q_{-,W},q)f(q)\,dq.$$

Combining the reduced-field transmission laws with the moved wall event then fixes the exterior areal derivative,

$$c_{1,r,W}=\frac{2q_{-,0,s}}{d}I_--\left(e-\frac{vV_1}{V_{0,s}}\right)\frac{2mw}{x^3d^2}$$
$$d=v-\beta_+$$
$$e=\rho-\frac{xp_W^2}{12}.$$

Here $w=c_{0,W}$ and $q_{-,0,s}=(\beta_{-}-v)/(1-x^2)$. The two terms are cap backscatter and the displacement pullback. At $x=0.55$, the primary 257-node field calculation gives $J_{-}=-1.8223975002$, a backscatter term -2.9736230746 and a displacement term -0.5062222284, hence $c_{1,r,W}=-3.4798453030$. The outgoing correction is therefore nonzero for this prepared pulse; the leading zero-outgoing boundary condition cannot be reused at this order.

Over nine common wall cuts, changing the field from 129 to 257 nodes changes $c_{1,r,W}$ by at most 1.73×10^{-6}. Radial order 8/16 changes it by 3.99×10^{-7} and ray order 8/16 by 3.26×10^{-3}. Separate 16/32 controls at 129 field nodes reduce these changes below 1.17×10^{-10} and 1.31×10^{-10} respectively. In the primary calculation, independent integrations in incoming label and radius agree in J_{-} within 5.59×10^{-11}. These are numerical sensitivity checks, not a rigorous total error enclosure.

The boundary samples provide a determined input for the exterior forced equation. The following calculation resolves its interpolation and propagates the exterior correction. No new physical preparation mechanism or nonzero finite-amplitude validity range is asserted.

13.14. The exterior scalar correction propagates on the computed geometry

Evidence: mass-moment comparison, clock-resolved boundary, forcing cache, scalar evolution and the conservative equation.

Pull the exterior areal coefficient back to the reference null coordinates: $y(U,V)=c_1(r_0(U,V),V)$. The same metric and first scalar jets determine the known flux F and density G in $y_{UV}+K_0y=F_U-G$. The original prescribed source has the determined areal-coordinate pullback $g_1=-H_{E,src}c_{0,r}|_{src}$. It is retained after the compact leading input; setting it to zero would change the preparation continuation.

Let $B_U=b_{out}$, where $b_{out}=r_{0,U}c_{1,r}|_W$ is the outgoing boundary derived above, and choose $B=0$ below the initial wall coordinate. Then $z=y-g_1-B$ has zero source and bottom data and $z_U|_W=0$. Its conservative evolution is

$$z_V = F(U,V) - F(U_0,V) - \int_{U_0}^{U} [K_0(z+g_1+B)+G]\,dU.$$

The fixed-U discretization clips inactive nodes to the moving wall. The terminal z state has no extra advection term, while the reconstructed wall derivative includes the lift transport:

$$\frac{dy_W}{dV} = g_1'(V) + z_V|_W + U_W'(V) b_{\mathrm{out}}(U_W(V)).$$

A cumulative pre-arrival cap-mass moment agrees with independent radial integration within 2.50×10^{-9} at the retained samples and makes the boundary evaluation economical. Its local Hermite lift initially exposes a clock interpolation error. Refining the exterior clock from 16 to 32 panels reduces the withheld boundary discrepancy from 1.73×10^{-6} to 5.75×10^{-8}, below the unchanged target. The lift uses 41 training knots and 40 withheld queries. The exterior forcing cache independently compares 17/33 time knots against 32 withheld times, including radial midpoint queries.

Four evolutions then hold the physical source, 129-node leading field and 257-knot reference geometry fixed:

Numerical comparison	Maximum y_W change	Maximum dy_W/dV change
17/33 forcing times, 129 correction nodes	1.755×10^{-3}	1.326×10^{-4}
65/129 correction nodes, 33 forcing times	6.984×10^{-6}	9.317×10^{-5}
Relative tolerance $2\times10^{-6}/2\times10^{-7}$, fine cache/grid	3.728×10^{-9}	3.956×10^{-8}

The finer sampled peak is 0.2679995064; the terminal coefficient is 0.2194607242. The cache-time change is 0.655% of that peak, the grid change 0.00261%, and the ODE change much smaller.

Source-value, source-rate and wall-flux defects vanish. Manufactured moving-wall solutions independently check the integration signs, source subtraction and material derivative. The four evolutions take 5.9375 process-CPU seconds and 950,926 scalar kernels using the saved inputs.

This is a formal exterior cubic scalar response for the declared preparation. Sampled refinement differences are not uniform bounds. The central readout additionally requires reflected cap forcing and the observer-clock pullback; no finite amplitude or remainder is fixed here.

13.15. Reflected cap transport gives the central scalar correction

Evidence: central derivation, radial comparison and incoming-ray comparison.

Write the dimensionless Jordan central proper time as $T=\tau_C/\ell$ and put $u=T-\operatorname{atanh} r$, $q=T+\operatorname{atanh} r$. The leading reduced field is $c_0=f(q)-f(u)$. Unlike the outgoing wall ray, an incoming ray to the centre can contain both profile terms. Its metric and forcing therefore use the full reflected field.

Write c_N for the central-clock null correction. Let $a(q)$ be the outgoing label of the wall event with incoming label q, and write $J(q)=c_{N,u}|_W$. Its wall value is

$$H(q) = y_W(V(q)) - C(q)f'(q)$$
$$C = \Delta C_{\text{fixed } V}.$$

The forced cap equation is $c_{N,uq}=-K_1c_0$. Differentiating the wall value gives $c_{N,q}|_W=H'-a'J$; integrating inward and imposing regular reflection at $u=q$ then yields

$$p_{1,C}(T) = 2H'(T) - 2a'(T)J(T) - I_C(T)$$
$$I_C(T) = \int_0^{x(T)} W_1(T-\operatorname{atanh} r, r)[f(T) - f(T-2\operatorname{atanh} r)]\,dr$$

$$W_1 = \frac{2\nu_E}{r^3} - \frac{p^2}{3} - \frac{p_T^2}{6(1-r^2)} + \frac{(1-r^2)p_r^2}{6}.$$

Here $x(T)$ is the radius of the wall event whose incoming label is T. The factor two follows from regular reflection. The wall derivative uses $y_{W,s}=V_{0,s}\,dy_W/dV$ and the differentiated optical constraint for C_s. These include the arrival-rate and arrival-shift contributions at fixed central time; adding another shift would count that pullback twice. The cap's cubic frame term cancels its central lapse pullback. This last cancellation is an algebraic check, not a second evolved solution.

The formal observable is

$$\frac{\ell\phi_C(T)}{\varepsilon} = p_0(T) + \alpha p_{1,C}(T) + O(\alpha^2)$$
$$p_0 = 2f'(T).$$

Five cuts lie strictly inside the retained arrival interval. With incoming radial and ray orders both 32, they give:

Central time T	Leading p_0	Cubic coefficient $p_{1,C}$
-0.54363418	*0*	*3.49×10^{-137}*
-0.46546305	*42.21372894*	*3395.219*
-0.42507962	*-0.265835912*	*-34413.869*
-0.38358521	*-40.14663419*	*4369.585*

Central time T	Leading p_0	Cubic coefficient $p_{1,C}$
-0.29640937	*0.00127167605*	*1454.355*

The 16/32 radial comparison changes the coefficient by at most 1.098×10^{-5}; the independent 16/32 incoming-ray comparison changes it by 5.285×10^{-5}. Passing the existing exterior cache, grid and ODE comparisons through the same cap algebra changes it by 1.155×10^{-4}, 1.667×10^{-4} and 5.958×10^{-8} respectively. These are separate sampled sensitivities, not a combined uncertainty enclosure. The leading field uses 129 nodes and the reference geometry 257 knots. Compatible saved cap and wall data are reused; no new wall, cap-clock or exterior evolution is performed. The radial pair takes 27.9375 process-CPU seconds and the ray successor 50.046875 seconds, each within its own 120-second and 30-million-kernel budget.

The quiet sample remains below 3.49×10^{-137}. Positive-radius centre samples retain the potential separately from its analytic $K_1=O(r^2)$ limit. The radial-32/ray-16 travel integral reproduces the previous cap calculation within 1.43×10^{-14}; changing to ray order 32 moves it by 5.21×10^{-10}. Manufactured controls check the optical derivative, integration sign, reflection factor and clock conversion independently of this physical run.

The coefficient is large for this preparation, particularly near the leading field's zero crossing. Its size neither selects a finite amplitude nor bounds the omitted term. The exact nonlinear equations now specify the next connection. The same action fixes the canonical Einstein potential and wall tension:

$$\widehat{V}_{\pm} = 3h_{\pm}^{2}\cosh^{4}(\Psi/\sqrt{6})$$
$$\sigma_E = \sigma\cosh^{3}(\Psi/\sqrt{6}).$$

An exact source-leg constraint admits a nonzero small-amplitude interval with positive radius and conformal factor. Extending that control through the bulk, moving interface and regular centre, with a quantitative remainder for this observable, remains open. The following calculation tests the coupled equations on the original window without assuming that estimate.

13.16. A nonlinear coupled window and its unresolved accuracy

Evidence: six-case evolution, source and runtime manifest, preceding transport comparison and figure record. The finite-amplitude owner specifies the equations, gauge, source and discrete representation.

The original compact preparation now drives a simultaneous cap, moving wall and exterior calculation at $\alpha=0.00012983127142551443$ and at its normalized zero-amplitude limit. The source label spans $0.4\le x\le0.4815042083$, corresponding to reference wall radii $0.45\le R\le0.55$. This window ends before the signal reaches the centre. The action, source, wall tension and reference gauge are held fixed; the amplitude is selected by the source-only bound, not an observed output. Each amplitude uses 16 cells with 256 steps, 32 cells with 256 steps and 32 cells with 512 steps, retaining 129 common history cuts.

The cap carries a reflected characteristic background

$$\chi_{\mathrm{can},0}(T,r) = F_{\mathrm{bg}}(T+\operatorname{atanh} r) - F_{\mathrm{bg}}(T-\operatorname{atanh} r)$$

and evolves corrections to its conservative cell integrals. Here $\chi_{can}=r\Psi/\sqrt{\alpha}$ is the normalized canonical reduced field; the transmitted invariant field c differs from it at finite amplitude. One cap primitive determines the common wall field. The full incoming correction remains evolved, and total fields, including all cross terms, source the cap metric. The C^2 quintic profile uses 65 alternating cuts of the previous zero-case wall history and its first jets. The 64 withheld cuts give maximum value and first-jet discrepancies 1.808×10^{-5} and 0.01439. Interpolation curvature is a computational choice, not an independent physical second derivative.

An independent characteristic pullback from each new zero-case wall trace tests cap propagation. Its 32-cell primitive discrepancy falls from *0.380945* to *0.000890490* with linear interpolation, or from *0.381936* to *0.000836671* with PCHIP. The remaining discrepancy is comparable to the *0.00133* interpolation sensitivity. Zero-case outgoing *p-q* remains exactly zero, and its spurious pre-arrival centre field is below 3.9×10^{-15}. These checks distinguish accurate incoming transport from an identity imposed by the cap primitive.

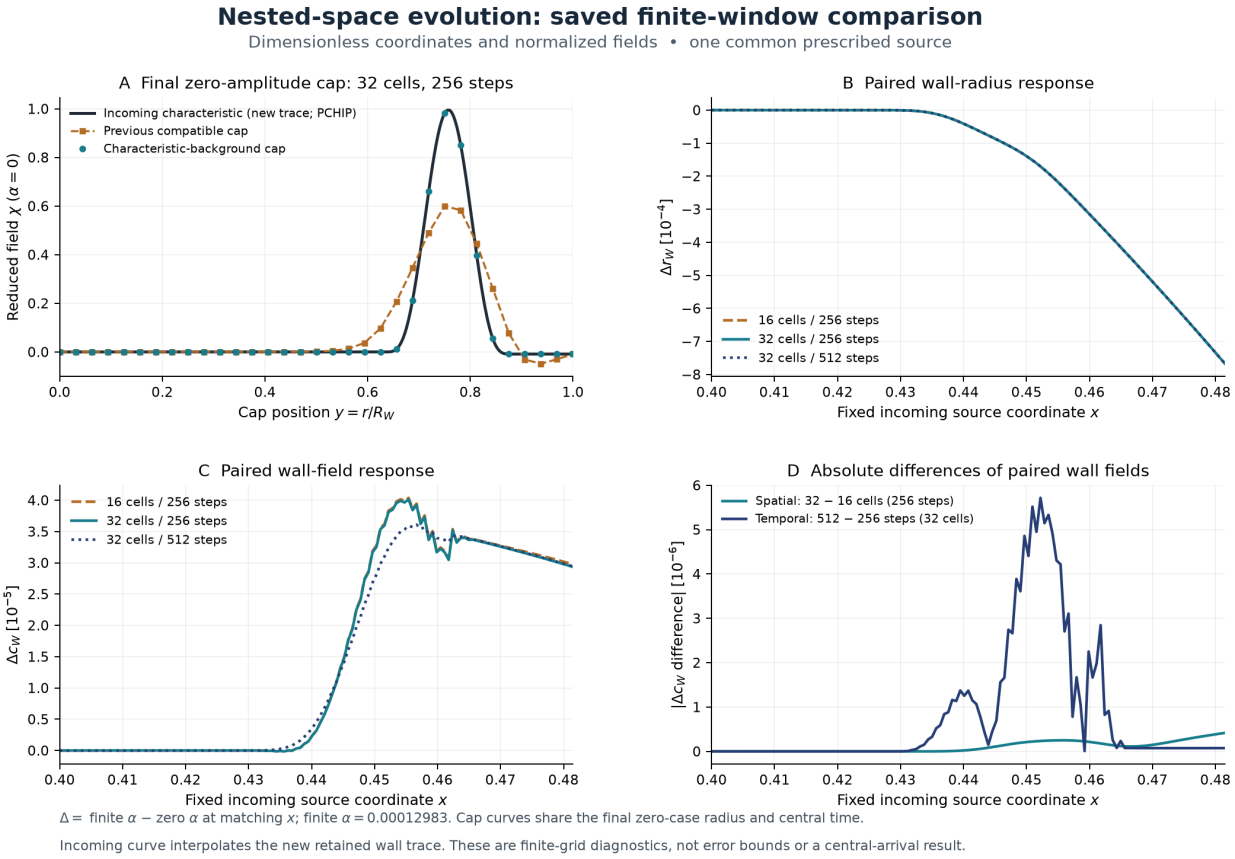


Figure 10. Saved finite-window comparisons. A: the previous and current zero-case cap primitives, with an independent incoming characteristic pullback from the new wall trace. B and C: finite-minus-zero wall radius and scalar histories. D: absolute spatial and temporal changes of the paired scalar response. The comparisons neither bound the continuum error nor extend to central arrival.

Define each response as the finite-minus-zero history at the same source label. In the table, spatial differences compare 16/32 cells at 256 steps; temporal differences compare 256/512 steps at 32 cells. Endpoint values use the 32-cell, 512-step pair. Here *L=log C*, *T* is the central coordinate clock, *s* is Einstein proper wall time and *c* is the normalized invariant wall scalar.

Quantity	Endpoint response	Maximum spatial difference	Maximum temporal difference
Radius *r*	-7.67297×10^{-4}	1.57484×10^{-7}	7.56646×10^{-9}
Log lapse *L*	-6.48536×10^{-3}	1.92145×10^{-6}	5.58556×10^{-8}
Central clock *T*	-1.27928×10^{-3}	8.00326×10^{-7}	1.10009×10^{-8}
Wall clock *s*	-1.19742×10^{-4}	4.54211×10^{-8}	6.59243×10^{-10}
Wall scalar *c*	2.94167×10^{-5}	4.18041×10^{-7}	5.71668×10^{-6}

Geometry and clock spatial differences are below 0.063% of their fine response peaks, and time differences below 0.001%. The scalar time difference is instead 15.83% of its fine peak 3.61049×10^{-5}. Its small endpoint time difference, 7.42×10^{-8}, therefore does not characterize the history. The finite

outgoing characteristic also changes by 12.41% of its fine peak under spatial refinement. Frozen cap radial order-4/8 comparisons change the wall mass coefficient by at most 3.00×10^{-8} and the lapse by 2.29×10^{-11}; this quadrature sensitivity does not explain the scalar time difference.

Independent exterior constraint maxima (E_U,E_x) decrease from $(7.88\times10^{-4},0.01312)$ to $(5.48\times10^{-4},0.008783)$, but remain nonzero. The source-array E_x defect remains 0.001303, and final 32-cell, 256-step wall kinematic defects in (r,L,c) are approximately $(-0.0003125,0.007739,-0.000119)$. Constructed scalar flux agreement near roundoff does not remove these independent derivative defects. Thus the integrated geometric responses are reproducible within the tested comparisons, while scalar time resolution, exterior constraints and wall slopes still require correction.

Creation and authentication use 228.9375 process-CPU seconds and 43,551,872 charged points within a shared 600-second, 100-million-point allowance. The manifest binds 66 inputs and producers. Authentication reconstructs saved diagnostics without source integration or time evolution. No continuum error enclosure, converged nonlinear spacetime, central-arrival waveform, formation mechanism or stability result follows from this window.

13.17. Profile-knot phase and a finer temporal comparison

Evidence: 128/256-step histories, 512-step comparison, source-bound manifest and figure record. The time-contribution derivation retains the background chain, physical wall flux and frame conversion separately.

The same spatial operators and physical inputs now use the three-stage SSPRK3 method at 128, 256 and 512 steps, always with 32 cells. The last comparison adds only the missing midpoint data: the original 513 rows are preserved exactly, and the same source snapshot supplies 512 new midpoints. Direct reference-gauge quadrature at orders 32 and 64 differs by at most 1.99×10^{-13} across its values and jets. The source geometry still uses its inherited Hermite approximation; its residuals are retained. Neither the physical preparation nor the 65-knot computational cap profile is refitted.

The latter profile is C^2, with possible jumps in its third derivative. An exact local model explains why the position of a knot inside a step matters. For a prescribed path with $B(x)=A(x-x_*)_+^3/6$ and $\theta=(x_*-x_0)/h\in[0,1]$, the SSPRK3-weighted background-chain defect is

$$\mathcal{R}_B = B(x_0+h) - B(x_0) - \tfrac{h}{6}\{B'(x_0) + 4B'(x_0+h/2) + B'(x_0+h)\}.$$

$$\mathcal{R}_B = \tfrac{Ah^3}{12}(1-2\theta)\min(\theta,1-\theta)^2.$$

It vanishes at an endpoint or midpoint and has largest magnitude $|A|h^3/324$ at phases $1/3$ and $2/3$. For a smooth increasing wall label $\zeta(x)$ crossing a profile knot, the leading coefficient is $A=[F_{bg}''']\zeta_x^3$. This local quadrature identity does not prescribe the coupled stage trajectory or identify its entire error. In particular, shifting the crossing phase on refinement can change the sign and size of a difference without a fixed observed convergence order.

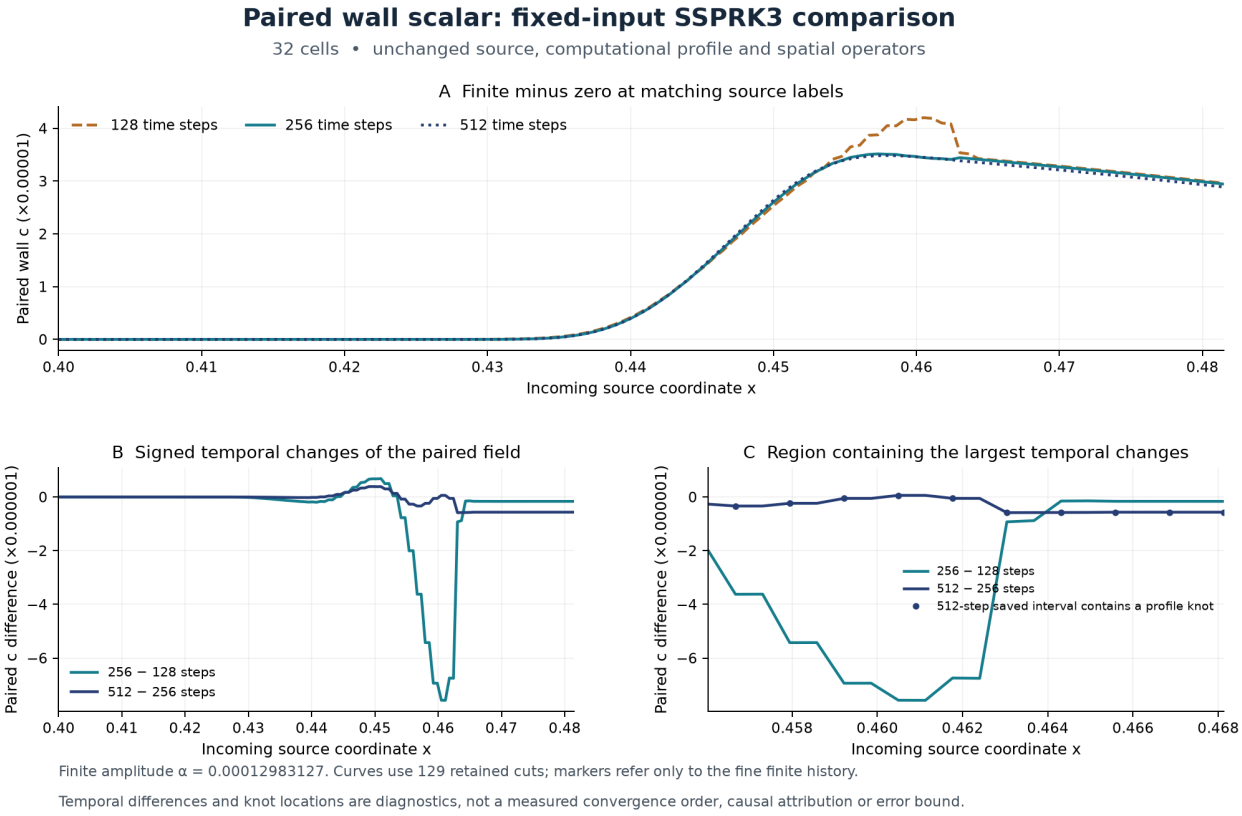


Figure 11. A: finite-minus-zero wall scalar at 128, 256 and 512 time steps. B: signed changes between successive histories. C: the region containing their largest changes; dots mark fine finite-history saved intervals containing a profile knot. These endpoint brackets do not reconstruct internal-stage paths or assign causation.

The paired scalar maximum time difference falls from 7.57345×10^{-6} for 128/256 steps to 5.84100×10^{-7} for 256/512 steps. The latter occurs at saved cut 99 and is 1.67616% of the fine peak 3.48476×10^{-5}. The difference ratio, 12.966, is a measured quotient, not a convergence order. The final scalar change is -5.66081×10^{-7}, so an endpoint alone does not characterize the history. All entries below use finite minus zero at common labels.

Quantity	512-step endpoint response	Maximum 256/512-step difference
Radius r	-7.67296891×10^{-4}	6.62412×10^{-11}
Log lapse L	-6.48536218×10^{-3}	6.54539×10^{-10}
Central coordinate clock T	-1.27927881×10^{-3}	1.38604×10^{-10}
Wall proper clock s	-1.19741847×10^{-4}	1.09942×10^{-11}
Wall scalar c	2.88661471×10^{-5}	5.84100×10^{-7}

The earlier SSPRK2 extrapolant $(4\Delta c_{512}-\Delta c_{256})/3$ agrees with the new 256-step history within 5.60381×10^{-8}, but with the new 512-step history only within 5.92862×10^{-7}. The closer earlier agreement therefore does not certify the true error. Independent finite constraint maxima remain approximately $(5.55\times10^{-4}, 0.008769)$ for (E_U, E_X). The unchanged computational profile's pre-support contribution also remains a separate representation issue: fine cap-integral leakage is about 1.66×10^{-7} even though the wall scalar leakage falls to 1.25×10^{-9}. Spatial defects, profile support, central arrival and a uniform finite-amplitude remainder remain open.

The two new histories and their validation use 174.90625 process-CPU seconds and 33,604,170 points, within 300 seconds and 50 million points. Their manifest binds 86 inputs and producers. A separate

authentication uses 1,418,330 points, reconstructs midpoint data and two final stages, and replays no source integration or trajectory.

13.18. A common exterior derivative and a mixed accuracy result

Evidence: fixed-input spatial comparison, same-grid operator comparison and source-bound manifest. The common-derivative construction specifies the discrete change and its limited consistency and stability controls.

The preceding 32/64-cell comparison at 512 SSPRK3 steps changes the paired wall scalar by at most 2.69431×10^{-7}, or 0.775% of the fine peak. This does not resolve the whole field: adjacent-summed cap integrals still differ by about 7% of the corresponding old 32-cell paired peak, and their physical radial cells move differently. The successor holds 64 cells, 512 steps, both amplitudes, source data, reference gauge and computational cap profile fixed. It replaces the first-order exterior departure derivative with a quadratic stencil. One derivative supplies both the radial first jets and transport on the moving coordinate grid. No boundary derivative is replaced by a value demanded by the wall law.

The independent constraints use saved field differences and analytic vacuum jets, rather than the evolution stencil. Write $E_a=r_{aa}-L_a\,r_a+r\Psi_a^2/2$ for $a=U,x$. The outgoing diagnostic retains the actual source curvature through $R_x=E_x-r_{src}E_{x,src}/r$. All entries below concern the finite-amplitude history on the same 8,001 interior diagnostic points.

Diagnostic	First-order derivative	Quadratic derivative
Maximum absolute E_U	3.53518×10^{-4}	3.64417×10^{-4}
Root-mean-square E_U	6.20959×10^{-5}	2.24521×10^{-5}
Maximum absolute R_x	4.93833×10^{-3}	3.21118×10^{-3}
Root-mean-square R_x	1.14164×10^{-3}	2.98901×10^{-4}

At the final wall event, the kinematic mismatches in (r,L,c) change from $(-1.54418\times10^{-4},3.75664\times10^{-3},-6.02063\times10^{-5})$ to $(4.70092\times10^{-6},-1.74363\times10^{-4},2.84633\times10^{-6})$. The maximum method change in the paired wall scalar is 1.81989×10^{-7}, or 0.524% of its new peak 3.47257×10^{-5}. Paired geometry and clock changes are each below 0.018% of their respective new peaks. These are sensitivities to the chosen derivative, not bounds on the continuum solution.

The improved average residual and wall slopes do not settle wall curvature. Independent backward-stencil estimates of the wall E_U peak increase from 4.87×10^{-4} and 4.49×10^{-4} to 6.67×10^{-4} and 7.73×10^{-4} at native width. The first estimate uses three nodes for first derivatives and four for second derivatives; the other uses five for both. Their maxima of the native-minus-double-width differences are 3.58×10^{-4} and 6.06×10^{-4}. The source Hermite curvature remains a one-sided trace at saved knots; its nonzero constraint is retained, not replaced by the source ODE target. The cap's characteristic-pullback discrepancy remains comparable to its interpolation sensitivity, and pre-support cap-integral leakage remains about 3.34×10^{-7}. The wall responses are comparatively stable under this method change, while local derivative accuracy remains unresolved. Neither a junction failure nor a converged coupled spacetime follows. Central arrival and a uniform finite-amplitude remainder remain open.

The two new histories and authentication use 256.25 process-CPU seconds and 55,398,966 charged points within 450 seconds and 80 million points. The manifest binds 104 files. Separate authentication uses 4,040,624 points and reconstructs saved fields, diagnostics and two final stages without source integration or trajectory replay.

13.19. A finite-amplitude onset solution

Evidence: exact reference gauge, quiet initial anchor, actual source bounds, and conditional tube receipt. The complete construction and continuation argument supplies the mathematical connection between those inputs and existence.

Finite-cell result. With the same action, signed material-wall laws, original compact C3 source and exact quiet initial data, the cap, wall and exterior admit a compatible regional solution on

$$x_L = \frac{475452113650795}{1125899906842624}$$
$$x_R = \frac{3809352257986645}{9007199254740992}$$

$$0 \le \alpha \le \frac{4789928473501569}{36893488147419103232} \simeq 1.2983 \times 10^{-4}.$$

Here x is the fixed affine source label. The cell is approximately *[0.4222863,0.4229231]*; it is not the complete earlier wall-radius window. Native fields are C3, native radius is C4, and the polar cap metric is C2. The centre remains a regular quiet de Sitter core. Local uniqueness holds in the stated regular class and fixed gauges.

The construction first integrates the outgoing mixed equations, then solves the existing wall ODE, transports incoming jets from their actual entry, and reconstructs each field from the same frozen forcing. This preserves field primitives and common wall values at every iteration. Exponential weights control the long exterior prefix; a cap strip based on its spacelike initial data has short characteristic lengths. Weighted derivative bounds and moving-entry estimates give a local contraction. The resulting wall laws and exact initial/source data propagate both incoming Einstein constraints.

Two ordered Volterra/transport derivative pairs give C2 first jets from the actual C3 source. Their finite bounds and continuity moduli remain uniform on the declared interval. The physical cap clock is constructed from the normalized radius-gradient flow. Its Jacobian stays nonzero; the transformed lapse is fixed by that flow and the quiet central clock. Restart slices keep their actual curved shape in the original null coordinates. Separate estimates for old radii and radii newly enclosed by the wall keep these slices inside the constructed domains.

On every initial subinterval, the same interval majorants improve all ten assumed departure bounds strictly; their largest ratio is below 0.516. The positive travel gap keeps the core quiet. A first exit through the field bounds, regularity, coordinates or quiet overlap is therefore excluded, and the positive source-label rate carries the solution to x_R.

For the bounds below, $f=\Psi/\sqrt{\alpha}$ denotes the normalized canonical scalar, with its continuous normalized limit at $\alpha=0$. The reduced canonical fields are $X=rf$ outside and $\chi=rf$ in the polar cap; these differ from the invariant reduced field $c=r\sinh(af)/a$ used in the native construction, where a is the nonnegative root of $6a^2=\alpha$. In the cap, $z=a\tanh r$, $\beta=Ae^{\delta}/(1-r^2)$ and $E^{\pm}=\chi_T/\beta\pm\chi_z$. The quantities J_r, J_L, J_X are integrated transverse-jet departures. All departures use the exact vacuum reference and the same fixed exterior wall and source labels. The resulting dimensionless bounds include:

Quantity	Strict upper bound
Integrated radius-jet departure J_r	5.662×10^{-9}
Integrated log-lapse-jet departure J_L	7.183×10^{-8}
Integrated canonical-field-jet departure J_X	1.474×10^{-10}
Absolute wall-to-central-clock departure ΔT	7.061×10^{-9}
Absolute incoming cap field E^+	0.002061
Absolute outgoing cap field E^-	3.639×10^{-19}

Quantity	Strict upper bound
Absolute cap reduced field χ	3.129×10^{-6}
Absolute exterior radius-jet departure $\Delta\, r_U$	1.451×10^{-6}
Absolute exterior log-lapse-jet departure $\Delta\, L_U$	4.224×10^{-5}
Absolute exterior canonical-field jet X_U	2.098×10^{-9}

The exact interval endpoints are retained in the receipt. That immutable record reports a conditional bound, not existence by itself; the separate construction and continuation argument supplies existence. Symbolic controls check the field primitives, wall tangents, differentiated equations and coordinate identities. No new physical trajectory is used in this proof. These bounds do not certify the saved numerical histories, a perturbative remainder, central arrival, formation or observational agreement.

13.20. Amplitude dependence on the onset cell

Evidence: source and comparison-chart regularity, coupled linear inverse, first quotient with one coordinate derivative, and second quotient and Taylor estimate. The source-bound record and conditional tube record retain the numerical input bounds; the linked derivation supplies the parameter theorem. Neither record evaluates its Taylor constant.

Amplitude result. On exactly the source cell and coupling interval of Section 13.19, the normalized solution has a continuous first amplitude derivative into C1 comparison jets and a continuous second amplitude derivative into C0 comparison jets. At the interval endpoints, derivatives are one-sided. The wall radius and anchored central clock are included.

The comparison uses the original source label x. The exterior has the fixed interval $0\le U\le W(x)$; the complete cap uses $r=r_c+\eta(R(x,\alpha)-r_c)$, $0\le\eta\le1$, with a fixed radius r_c in the quiet core. This keeps all radii enclosed by the wall, including those already present at the initial event. Each solution retains its own clock $T(x,\alpha)$ and the actual coordinate Jacobians. In particular, T labels the wall's polar slice by the anchored central proper clock; it is not the arrival time of a later signal at the centre.

Let $U(\alpha)$ collect the normalized comparison state and write $U_1=\partial_\alpha U$ and $U_2=\partial_\alpha{}^2U$. Exact finite differences first give Lipschitz dependence into C0. Uniform spatial C2 bounds and continuity moduli then give continuity into C2 by second differences. The first parameter quotients converge into C1 in the order outgoing exterior derivative, cap gradients, incoming exterior derivative. The cap's radial constraints express the radial derivatives of its propagation coefficients without extra field-gradient unknowns. Actual inherited initial data and the wall corner identity are retained at each restart.

For a cap transport equation $(\partial_x-\sigma\partial_\eta)w=R$, its second variation has the same coupled linear operator as the first. Put $v_1=(U_1,1)$, including variation of the explicit parameter. The known second-order forcing is

$$2D\sigma[v_1]\,\partial_\eta w_1 + D^2\sigma[v_1,v_1]\,\partial_\eta w + D^2\mathcal{R}[v_1,v_1].$$

The factor two comes from varying the transport speed and the field gradient separately. Convergence of the first quotients in C1 controls this forcing in C0. The bounded coupled inverse and a finite sequence of compatible time slabs give convergence of the actual second quotients, not just a formal differentiated equation. The exterior, radial mass, lapse, wall and clock equations participate in the same limit. No fourth source derivative is used, and no C1-valued second derivative of all jets is asserted.

For a specified normalized state component, wall radius or anchored clock O, define the output-specific finite constant

$$M_{2,O} = \sup_{0\le\alpha\le\alpha_{\max}} \|\partial_\alpha^2 O(\alpha)\|_{C^0}.$$

Twice integrating in amplitude gives

$$\|O(\alpha) - O(0) - \alpha O_1(0)\|_{C^0} \le \tfrac{1}{2} M_{2,O} \alpha^2.$$

This is an analytic quadratic remainder on the onset domain. The general state constant is finite; Section 13.21 supplies explicit remainder bounds for selected geometric outputs by a positive comparison. It also identifies their first coefficients by matching the source, gauge and quiet initial data. Errors in the older numerical histories require their own bounds. The normalized fields have the stated amplitude derivatives. The physical scalar $\Psi=\sqrt{\alpha} f$ is not asserted to be twice differentiable in α at zero. The quiet core, original source and coupling interval are unchanged. Continuation through central arrival and an error bound for the observer waveform remain separate steps.

13.21. Uniform geometric remainders and signed endpoint response

Evidence: first geometric coefficients, geometric, canonical scalar and normalized cap first-difference bounds, quadratic source remainders, complete geometric remainder, and Jordan material-radius conversion. The comparison and frame derivation connects these compact certificates.

Keep exactly the source cell, action, quiet data and coupling interval of Section 13.19. All quantities here are dimensionless. The outputs are the Einstein areal radius R_E, the material Jordan areal radius R_J, and the wall's centre-anchored polar-slice label T_W. Their first coefficients are the actual derivatives of the constructed solution at $\alpha=0$. They coincide with the continuum source, mass, lapse, wall and clock functionals after the source-label event shift is included. This identification does not assign an error bound to a saved numerical history.

Uniform comparison. For each retained geometric component G, the calculation supplies an explicit constant C_G such that

$$\|G(\alpha) - G(0) - \alpha G_1(0)\|_{C^0} \le \alpha^2 C_G.$$

The radius estimate covers the retained exterior, including the wall; the clock and material-radius estimates cover their wall histories on this cell. The source supplies four quadratic remainders. Uniform first differences of the canonical scalar and normalized cap mass and lapse bound the remaining nonlinear terms. Half second derivatives of the exact bulk and signed wall maps are enclosed over the full certified state box, with the radius-kernel cross term and scalar kinetic product retained.

After division by α^2, the three coupled geometric bounds obey $y \le My + f_2$, with the same positive matrix as the first-difference comparison. Its weighted norm is below 0.146, so $y \le (I-M)^{-1} f_2$. The remaining outputs follow from its dependent rows. This uses one rational three-by-three solve and interval algebra, with no new field evolution or grid scan.

Material frame and event identification. With the normalized canonical field $f=X/R_E$ of Section 13.19, the exact conversion is

$$R_J = R_E \cosh\left(\sqrt{\alpha/6}\, f\right)$$
$$\mathcal{F} = \operatorname{sech}^2\left(\sqrt{\alpha/6}\, f\right) > 0.$$

The invariant Jordan trace $c=R_E\, f\sinh c(\sqrt{\alpha/6}\, f)$ is a different variable at finite amplitude. For a fixed target α, join $(0,R_0,X_0)$ to $(\alpha,R_\alpha,X_\alpha)$ by an affine segment. Its directions are the actual difference quotients, bounded by the certified first-difference constants. At the expansion base the frame map has state partials one and zero, and explicit alpha derivative $X_0^2/(12R_0)$. Therefore the Einstein radius remainder enters with coefficient one; the frame half-Hessian adds at most 1.604501×10^{-23} to C_{R_E}. The conformal factor has certified lower bound 0.9999999999999999 on the whole comparison.

The first Jordan coefficient at fixed source label is

$$R_{J,1}|_x = \rho + 2a_{\mathrm{ref}} v(v - \beta_+)\eta_{\mathrm{emb}}$$
$$a_{\mathrm{ref}} = 4/5.$$

Here ρ is the Jordan radius coefficient at fixed wall proper time, v is the reference proper-time radial velocity, and η_{emb} is the exterior embedding coefficient defined by $V_W=V_0(1+\alpha\eta_{emb})+O(\alpha^2)$. Its interval includes both signs and is retained in the conversion; η_{emb} is distinct from the cap chart coordinate of Section 13.20.

Endpoint result. The following are outward-rounded bounds. The last column gives the shift from the vacuum reference at x_R and $\alpha=\alpha_{\max}$, divided by 10^{-18}.

Output	Upper bound for C_G	Endpoint shift / 10^{-18}
Einstein wall radius R_E	6.067430×10^{-17}	[-2.138954,-1.008091]
Material wall radius R_J	6.067431×10^{-17}	[-1.235768,-0.09856682]
Anchored wall clock T_W	1.145597×10^{-15}	[-2.635641,-1.410894]

For a first-coefficient upper endpoint $U_G<0$, the criterion $\alpha_{\max}C_G<-U_G$ makes the sign strict at every positive admitted amplitude. The exact ratio is below 1.038×10^{-5} for the material radius and 1.369×10^{-5} for the clock. Hence

$$R_J(\alpha, x_R) - R_J(0, x_R) < 0$$
$$T_W(\alpha, x_R) - T_W(0, x_R) < 0 \quad (0 < \alpha \le \alpha_{\max}).$$

The clock retains the same Einstein-polar slicing and equals physical Jordan central proper time on the quiet core. Its value assigned to the wall event is neither wall proper time nor the arrival time of a received central signal. The reference wall is expanding: these inequalities are sourced differences at the same input label, not absolute contraction or an evaporation law. The preparation is still prescribed characteristic data; no physical emitter, SI scale or observational calibration is selected.

The geometric and material-frame certificates retain exact rational endpoints, producer commits and input hashes. Seventeen targeted controls, separate rational audits and independent Grok derivation/code reviews check the identities, finite boxes, matrix closure, frame conversion and strict inequalities. The audits authenticate the saved results without rerunning a field solution. Continuation to central arrival, numerical trajectory error, preparation selection and observational comparison retain their separate requirements.

14. A separate finite model with geometric feedback

Evidence: focused companion snapshot, finite episode, local response, and memory controls.

The finite companion supplies another part of the programme: fields transfer between selected regions while the same declared action evolves the geometry. It uses a periodic radial chart, positive chart variables and antiperiodic spinors. In the present signature convention its metric is

$$ds^2 = r^2[-L^2dt^2 + Q^2(dx + \beta\, dt)^2 + d\Omega^2].$$

The physical lapse is rL and radial scale is rQ. Conformal gauge has $L=Q$, $\beta=0$; the reduced field Hamiltonian is $H=\sigma_2 P+\kappa\, Q\sigma_1$. Here $\kappa=1$ labels the retained angular block and is independent of the wall-tension parameter used in the atlas. Six prepared Gaussian fermion columns determine

$$C = \Phi\,\mathrm{diag}(0.75, 0.75, 0.5, 0.5, 0.25, 0.25)\Phi^{\dagger}$$
$$\Phi^{\dagger}\Phi = I_6.$$

This is a finite admissible state with $0 \le C \le I$ and $\mathrm{Tr}\, C = 3$. Unoccupied retained bands are part of the preparation, not a derived physical vacuum. The action coefficients, magnetic sector, clock and cutoff are recorded inputs.

Four matched discretizations, using 256/512 field modes and time steps 0.001/0.0005, reach coordinate $T = 0.3$ without resetting the state. The reported transfer and curvature comparisons have sub-percent refinement indicators in their stated observables. The finite evolution supplies a same-action example, not a continuum error theorem, an eternal engine or the creation of new universes. Collars of this periodic geometry are regions of one evolving model.

Its local reduction keeps the original first two mode columns and eliminates their Hilbert-space complement. That complement includes modes with support inside and outside the spatial observation window. Consequently it is not the parent end of Section 10, and its Hamiltonian cannot be transferred to the wall model without a field-theoretic identification.

15. Memory is a consequence of elimination, not an added reservoir

For a time-dependent Hermitian block Hamiltonian with retained block $A(t)$, coupling $B(t)$ and complementary block $D(t)$, let $U_E(t,s)$ propagate D. Eliminating the complementary amplitude gives

$$y(t) = U_E(t, t_*)y_* - i\int_{t_*}^{t} U_E(t, s)B^{\dagger}(s)x(s)\,ds$$

$$\dot{x}(t) = -iA(t)x(t) - iB(t)U_E(t, t_*)y_* - \int_{t_*}^{t} K(t, s)x(s)\,ds$$

$$K(t, s) = B(t)U_E(t, s)B^{\dagger}(s).$$

The memory kernel, actual initial complementary drive and initial cross correlations all matter. If R_A, R_E denote the corresponding solution maps, the retained covariance is

$$C_A = R_A C_{AA} R_A^{\dagger} + R_E C_{EE} R_E^{\dagger} + R_A C_{AE} R_E^{\dagger} + R_E C_{EA} R_A^{\dagger}.$$

The companion uses the actual evolved state at $t_* = 0.2$, then compares the reduced and full histories through $t = 0.3$. The selected occupations change from approximately $(0.10557448, 0.10557464)$ to $(0.03214496, 0.03226900)$. The maximum reduced/full occupation discrepancy is 6.35×10^{-6}, compared with a physical-model change of about 0.07343.

Conditional omission control	Maximum occupation effect	Its separate numerical reference discrepancy
Omit the memory term	0.0168185	2.20×10^{-6}
Omit the actual initial complementary drive	0.0485412	1.66×10^{-5}
Omit initial cross correlations	0.0920614	1.23×10^{-5}

These controls hold the generated geometry fixed. They are not separately coupled physical evolutions; in particular, deleting the memory term is a diagnostic alteration, not a new Hermitian theory. They

demonstrate why an interior description must retain the appropriate history and state information. The next bridge to the atlas would derive such a regional Hamiltonian and state from a particular matched spacetime and field, rather than labeling these finite modes as universes.

VI. What the map can ask next

16. Constants, spectral scales and physical inheritance

Open physical target. A shared normalized operator can preserve dimensionless spectral relationships while regional clocks and absolute frequency units differ. To turn this into a statement about constants, the attachment must specify clocks, rods, fields and the transition of their dimensionless parameters. Useful comparisons include α, mass ratios, and $Gm^2/(\hbar\, c)$. A numerical change caused solely by unit rescaling is not a changing law.

Successive inheritance maps also need compatible output and input spaces: the first transition must supply data on which the next law is defined. Drawing another nested sphere does not establish this composition.

The atlas wall parameters $H\ell$, M/ℓ and $\kappa\ell$ are declared inputs, not values inherited from a parent calculation. Physical inheritance might preserve a law while changing states, select among admissible states, or alter an effective environment. These possibilities need different maps and should remain distinguishable.

The foundation also contains a scalar-link construction whose invariant sector has the ordinary Dirac relation $E^2=|p|^2+\Phi^2$. It is a useful benchmark relating a coupling to a gap. It does not derive the observed particle species, nuclear masses or their interactions. Particles are not assumed to be little universes. A particle interpretation requires the appropriate field representation, observables and dynamics, not resemblance between a spectral scale and a cosmological radius.

17. Wavelength, preparation and the role of light speed

The derived scales

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}$$
$$t_P = \frac{\ell_P}{c}$$

are defined combinations of constants, as tabulated by NIST CODATA. Dimensional construction alone does not establish a smallest physical wavelength or an indivisible spatial cell. A claimed minimum would need a model of probes, observables and gravitational backreaction. Minimal-length scenarios themselves distinguish limits on resolution, distance products and other quantities; see Hossenfelder, 1203.6191v2, Sections 5.3–6.

Absolute zero refers to thermodynamic temperature and the limiting thermal state, not to an endpoint of allowed wavelengths; see NIST's temperature definition. A relation between a regional temperature and a spectral scale requires a specified state and Hamiltonian. Changing a temperature unit supplies neither.

A horizon is not automatically a reflecting cavity wall or a prohibition on longer-wavelength modes. A particle-like exterior description likewise requires an additional quantum state and measurement model; exterior mass, spin and charge alone do not establish that description. Quantum observables can have continuous as well as discrete spectra. A detector event or a wave crest is not a general definition of a particle.

Separately, $T=0$ in an evolution record names a prepared state and a clock origin. It is not the short-wavelength endpoint of a spectrum. An ultraviolet cutoff, an initial time and a spatial boundary have different mathematical roles.

A local freely falling observer's null-cone speed, coordinate propagation rate, group velocity, phase velocity and a reduced response delay are also different quantities. A varying coordinate speed near a horizon is not evidence that the local causal constant changes. Claims about different regional *c*'s require an operational, dimensionless comparison and a consistent theory. Ellis and Uzan, gr-qc/0305099v2 explain why the distinct roles often assigned the symbol *c* should be separated. The present atlas uses conventional local null cones; its clock conversion and delayed response do not change that local law.

18. Dark gravitational effects require a stress and an observable

Open physical target. The unexplained gravitational and expansion phenomena motivating the project must be investigated after an exterior model supplies consequences. A prescribed fraction, including “95%,” is not inserted as an exterior abundance or a target that chooses the map.

Clustering and lensing require a stress or effective gravitational response with the right spatial, temporal and scale dependence. Accelerated expansion requires a separate background and perturbation analysis. Antimatter and clock arrows require still other data. The calibration scalar pulse has no calculated energy flux or backreaction and leaves no persistent central field in its certified interval. It supplies none of these dark components.

The foundation's Schur response and the library's shadow-matter and brane examples remain conditional starting points. A successful next connection would carry an actual source/state through the admitted geometry into a gauge-invariant gravitational observable. It would then either distinguish that exterior from alternatives or demonstrate a stated degeneracy. Fitting a freely chosen incident waveform cannot supply that discrimination.

19. Matter, antimatter and orientation

Open physical target. “Inside” and “outside” are geometric labels. Charge conjugation transforms fields and their gauge representations. Changing radial orientation or the sign of a chart coordinate does not, by itself, convert matter into antimatter. A proposed boundary identification must say which field components and charges it matches and whether the source, state and conserved currents permit it.

A real wave's sign change is not charge conjugation. CPT symmetry is also distinct from time reversal alone; neither supplies an entropy arrow.

The library preserves distinct CPT, black-mirror and torsion-production proposals. They cannot be combined by assigning one side of every sphere to antimatter. A useful next test would choose one actual attachment, write its field transformation and determine whether it selects a charge asymmetry from the stated initial data. The current real scalar has no particle–antiparticle charge degree of freedom and cannot settle this.

20. Time, recycling and the next generation

Open physical target. A continuing child needs a domain beyond a single initial slice and a source whose evolution remains meaningful there. The present wall reference has a singular past; the gluing theorem supplies initial data only. Neither is a proof that an astrophysical collapsing black hole forms the mapped cap, or that the cap is stable under disturbances. Regular background curvature and perturbative stability are separate tests.

Proper times can be compared along specified worldlines with a matching convention. Cross-universe ages cannot be read from unrelated coordinate times or unrecorded clock origins. Expansion, entropy increase, causal orientation and an observer's records may all participate in an arrow of time, but a clock conversion alone establishes none of those thermodynamic claims.

Open physical target. Finite-depth inheritance and periodic regional transfer do not prove cosmic recycling or an eternal source. A new generation requires an identified transition, source/state transfer and entropy accounting. Multiple holes motivate multiple possible offspring; they do not require equal children or communication between siblings.

Known branching mathematics should be borrowed. In the finite-type illustration used by Altenberg, 1302.1293v2, an offspring law determines a mean matrix N_{ij}; conditional expected populations evolve by N^n. Its long-time conclusions require the stated branching and spectral hypotheses. The unresolved physics is the transition law and its admissible parameter/state inheritance. Choosing that law to favor an intended cosmic population would assume the result. No infinite-generation computation is needed before one such transition has been established.

Conformal cyclic cosmology remains a separate possible continuation, with its own conformal matching and matter assumptions. It is not identified with black-hole offspring or communicating mouths by a scale analogy.

21. Measurements that could decide between maps

Open physical target. A discriminating prediction must survive changes of units, chart labels and unspecified initial data. The protected vacuum interval proves a finite geometric degeneracy. The scalar inverse in Section 13.3 determines the preparation for a fixed geometry and supplies preparation-class exclusions. It does not establish equivalence between two physical exterior models. Both results identify what information the next observer comparison must supply.

The next admissible calculations therefore have concrete decisions:

Missing connection	Finite decision	What it would unlock
Onset existence to a controlled observer response	Can the finite-amplitude solution be extended through central arrival with a controlled perturbative remainder?	A finite-amplitude prediction for the interior observer
Preparation-restricted observer inference	Does the source reconstructed from a measured trace satisfy an independently justified preparation law in each candidate?	A discriminating response-range test or an explicit equivalence class
Alternative coupling or source	Does a fixed minimal-coupling or sourced model leave a distinguishable tail?	A comparison of interior response mechanisms
Physical regional Hamiltonian	Does a chosen matched field yield the required reduction and initial covariance?	Memory with an identified geometric exterior
Stability or formation of an attachment	Do fixed-source disturbances or collapse data remain in the claimed domain?	A stronger cosmological interpretation
Dimensionless inheritance	Which transition quantities are preserved or changed?	A meaningful comparison of constants across regions

These are candidates, not a compulsory queue. Each proposed calculation must state what is borrowed, what is missing, what it decides, what it unlocks, its finite domain and budget, and the outcome that stops or redirects it. The existing admission owner keeps that rule. A failed attachment remains visible, while another geometric route can proceed under its own assumptions.

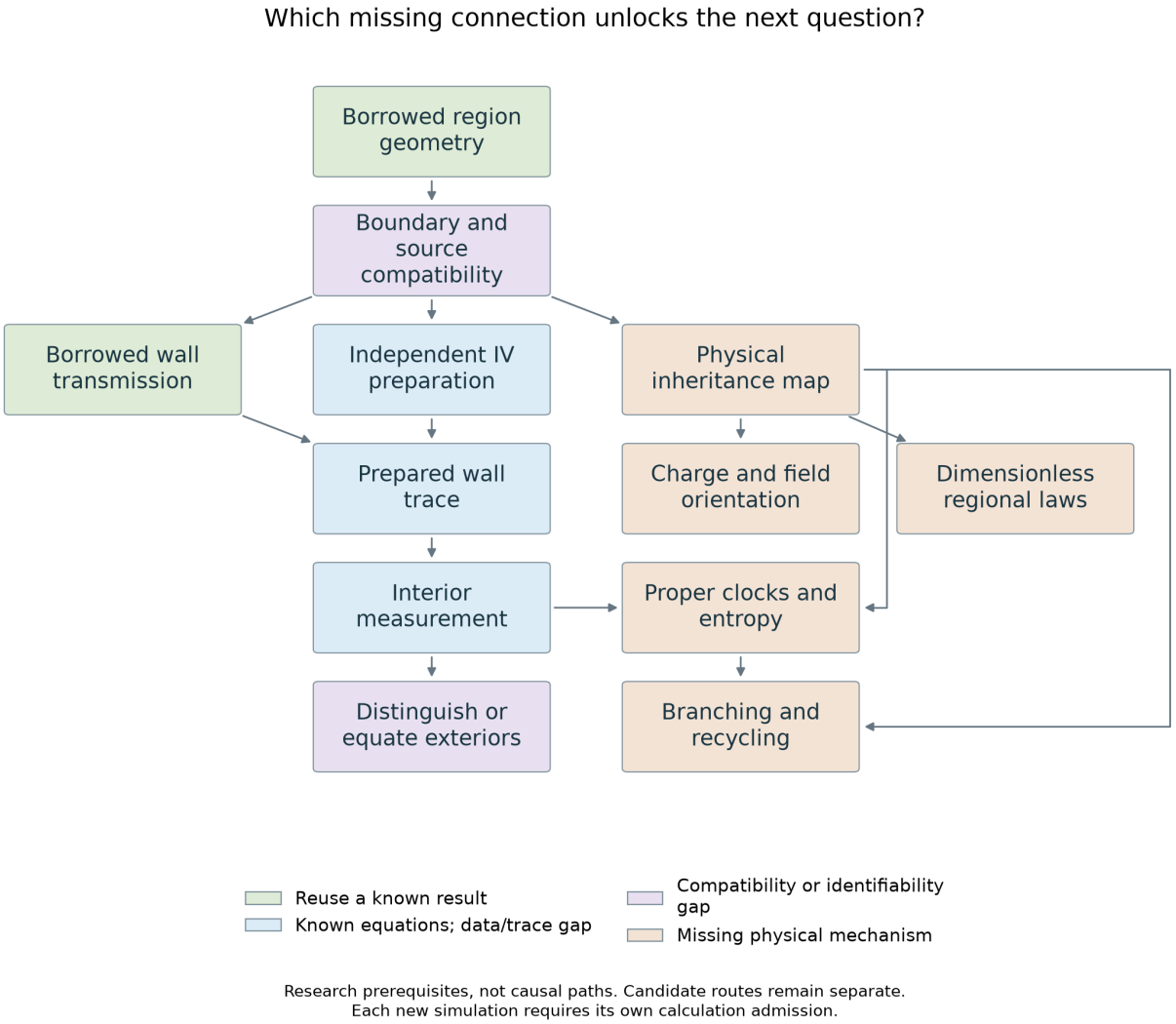


Figure 12. Research dependencies, distinct from spacetime causal arrows. The v8 programme record names prerequisites, decisions, budgets and stop conditions for eight themes. Each calculation retains its own domain; arrows do not declare questions solved.

The current result is a continuous mathematical story at a deliberately finite scale: an exterior can be attached on a sourced slice; an isolated completion determines observer access; a specified field state gives an interior waveform; independent finite exterior data now predict a pulse and a positive finite scattering tail; and finite regional reduction explains how unresolved degrees of freedom can retain memory in another fully stated model. The new inverse shows how a complete finite signal selects its preparation within a fixed geometry, and why physical source restrictions must accompany any comparison of maps. The quadratic stress check now connects that preparation to compatible local wall conditions. Its characteristic constraint requires a nonzero initial radius correction under the declared corner and null normalization; transverse constraint transport then supplies an invariant geometric mass coefficient on that same surface. The vanishing normal-stress jump adds a necessary condition on their later wall matching. Known field traces also determine an analytic exterior-side wall-mass functional without a new field solve. Subtracting the cap vacuum at the actual radius now supplies the other mass functional and the quiet-anchored wall displacement equation. The temporal and normal conditions agree with their common mass transport. The prepared field now drives an evolved exterior radius and lapse on a finite characteristic rectangle, with mass-transport and null-constraint checks. The retained-wedge evaluation now compares independent radial and timelike mass paths, computes the two-sided wall displacement and fixes its exterior clock correction. Radial cap reconstruction now also fixes the central proper clock and a corrected null-arrival map, with independent cap-mass transport and numerical controls. The formal exterior scalar correction is now propagated with independent numerical

comparisons. Reflected cap transport now also reconstructs its central coefficient at five fixed proper times. A nonlinear cap, wall and exterior calculation now also completes the original pre-arrival window. Its six-case comparison substantially reduces leading cap transport error and resolves integrated geometry and clock changes within the tested refinements. Scalar time sensitivity and independent constraint and wall-slope defects prevent a converged-solution claim. With the source and spatial operators fixed, SSPRK3 temporal refinement reduces the paired scalar change to 1.68% of its fine peak; computational profile-knot phase prevents interpreting a single difference quotient as an observed order. Spatial constraints and the profile's support defect remain separate. A fixed-resolution change to a common quadratic exterior derivative reduces average constraint residuals and the final wall kinematic mismatch; its paired scalar change is 0.524% of the new peak. The slightly larger incoming-constraint maximum and worse, width-sensitive wall-curvature estimates keep local accuracy unresolved. Separately, Section 13.19 establishes a finite-amplitude onset solution and its certified departure bounds. Section 13.20 establishes its first and second amplitude derivatives. Section 13.21 identifies the first geometric coefficients, supplies uniform quadratic remainders and proves negative material-radius and anchored-clock corrections at the onset endpoint throughout the admitted positive-amplitude range. Errors in the older numerical histories, continuation through central arrival and the observer waveform remain separate open questions. The remaining work is to connect those statements physically where their domains presently differ.

Appendix A. Evidence, attribution and preserved mathematics

This living manuscript draws from source-bound atlas records without changing their historical bytes. Record residuals and refinement comparisons are reported at their declared scope; floating-point quadrature agreement is not a rigorous error bound. The overview producer reads existing records and renders their relationships without launching a new physics calculation.

Result family	Owning derivation and preserved evidence
Finite inheritance and nested reduction	Proof; snapshot
Two-hole parent and radiation attachment screen	Atlas v2; manifest
Signed moving wall and gluing literature	Atlas v3; manifest
Sourced slice and vacuum observer domain	Atlas v4; manifest
Isolated causal completion and clocks	Atlas v5; manifest
Classical incoming state and scalar transfer	Atlas v6; manifest
Finite exterior preparation and research dependencies	Preparation; programme; v7 manifest
Evaluated exterior preparation and finite tail	Response; v8 manifest
Fixed-model source inverse and preparation exclusions	Proof; record
Conformal probe surface stress and tangential flux	Derivation; record
Characteristic gravitational initialization	Constraint and frame check; record
Geometric mass on the prepared null surface	Mixed geometry and stress checks; record
Conformal scalar normal-force compatibility	Projection and force sign; record
Exterior wall-mass functional	Both fluxes and full frame conversion; record
Cap mass and wall displacement functional	Vacuum subtraction and junction closure; record
Smooth exterior metric response	Characteristic evolution and radial checks; metric, field and derivative comparisons
Moving wall and exterior clock	Mass paths, displacement and clock; field, radial and clock-time comparisons
Smooth cap and central null clock	Regular centre, metric and arrival map; field, radial and further ray comparisons

Result family	Owning derivation and preserved evidence
Exterior cubic scalar correction	Conservative evolution; clock lift, forcing cache and propagated response
Central cubic scalar correction	Reflected transport and clock pullback; radial comparison and incoming-ray comparison
Nonlinear coupled pre-arrival window	Exact model and characteristic background; six-case result and manifest; scalar time sensitivity and pointwise constraints remain unresolved
Fixed-input SSPRK3 temporal refinement	Profile-knot and chain identities; 128/256 histories, 512 comparison and manifest; finite temporal differences, without an error enclosure
Same-grid exterior derivative comparison	Common derivative and independent diagnostics; 64-cell baseline, operator comparison and manifest; smaller average residuals and wall kinematic mismatch, with unresolved constraint maxima and wall curvature
Finite-amplitude onset solution and parameter dependence	Construction and continuation, parameter theorem; original source bound and strict tube
Uniform geometric remainder and material-radius sign	Full comparison, frame and clock interpretation; geometric certificate and material-radius certificate
Finite spherical transfer, feedback and memory	Companion; frozen build/evidence
Foundation boundary, spectral and source constructions	Archived v0.26.0; research inventory

The archived foundation retains, among other strands, its full-spinor boundary maps, scalar-link Dirac benchmark, source and work ledgers, curved spectral constructions and explicitly limited black-universe benchmarks. Their theory, state and domain remain with their owning notes. An RN pressure diagnostic is not promoted to mechanical balance, a reached inner horizon is not promoted to central regularity, and stopped numerical runs without completed summaries supply no new physical conclusion. These distinctions preserve useful work without making historical routes mandatory.

The arXiv library records paper versions, reading depth, reusable equations, assumptions and unresolved connections. It includes the sources for the supplementary multiple-daughter, multi-mouth, CPT, torsion and regular-interior branches. A bounded search cannot certify that no one has already closed an apparent gap. The paper guide locates the unchanged atlas working drafts and the focused companion. The author's responsibility includes the interpretations and synthesis here; borrowed results retain their source attribution and are not presented as new laws.